

stat_cot_observational_efficiency

Specialization: v1

TL;DR. Smooth conditional-optimal-transport inference already includes the wavelet stability and randomized marginal-sample asymptotic normality of Lin–Gao–Blanchet–Glynn. For the sharp lower endpoint on mean squared individual treatment effects, this note establishes three different observational results: the covariate-composition plus inverse-propensity canonical gradient under ignorability and overlap; the semiparametric information lower bound together with a propensity-corrected cross-fitted one-step estimator that attains it; and uniform studentized normality and Wald coverage on admissible positive-variance collections, including $\sigma_n \rightarrow 0$, under $\sqrt{n} \delta_n / \sigma_n \rightarrow 0$, $r_{S,n} / \sigma_n \rightarrow 0$, and $\sqrt{n} \sigma_n \rightarrow \infty$. At equality, the first-order pivot fails and the unregularized full-tangent Gaussian spectral route is obstructed by a noncompact, non-Hilbert–Schmidt Hessian; an honest full-parent adaptive procedure using controlled regularization or a different calibration remains open.

1 Project justification

Gap. Lin–Gao–Blanchet–Glynn already provide smooth-density wavelet conditional-optimal-transport stability and pointwise asymptotic normality for randomized marginal samples. Their theorem does not provide the observational tangent-space gradient with both covariate-composition and inverse-propensity terms, an information lower bound with an attaining observational estimator, or uniform studentized coverage along admissible sequences with $\sigma_n \rightarrow 0$. Other conditional-optimal-transport work supplies covariate-aware identification, observational direct-value rates, or AIPW inference for fitted dual bounds, but not this three-part efficiency-and-coverage result for the sharp smooth primal endpoint. The spectral existence criterion is settled; the remaining inference gap is a full-parent adaptive confidence procedure outside the ruled-out unregularized full-tangent Gaussian route.

Niche. The note treats the sharp conditional-coupling endpoint as an identified causal estimand observed through one i.i.d. observational sample. Relative to randomized smooth COT asymptotic normality, it derives the propensity-corrected observed-data canonical gradient. Relative to observational direct-COT value estimation and dual-bound AIPW inference, it proves the information lower bound and constructs a cross-fitted one-step estimator attaining it. Relative to positive-variance pointwise Gaussian theory, it retains the shrinking σ_n scale in the variance expansion and proves uniform studentized validity under explicit relative-rate conditions.

Fill. The results give the observed-data canonical gradient and efficiency bound, the matching information lower bound, an attaining cross-fitted estimator, the exact signed transport and propensity bias decomposition, a scale-aware variance error of order $\sigma_n r_{S,n} + r_{S,n}^2 + (\sigma_n + r_{S,n}) / \sqrt{n}$, uniform normality and Wald coverage under the three displayed relative-rate conditions, and explicit Hölder phase conditions. These conclusions complement rather than replace earlier smooth COT wavelet inference, observational direct-value estimation, and dual-bound AIPW inference. The note also proves a route-specific equality obstruction: the full-tangent conditional Hessian can contain an infinite-dimensional identity compression and hence be noncompact and non-Hilbert–Schmidt, so uniform unregularized Gaussian spectral calibration is unavailable without covariate-direction regularization or another derivative-aware method.

2 Related work

Lin, Gao, Blanchet, and Glynn’s 2026 smooth-density paper supplies wavelet conditional-optimal-transport stability and a pointwise asymptotic-normality result for randomized marginal samples; it does not supply the observational covariate-composition plus inverse-propensity canonical gradient, an observed-data information lower bound with an attaining cross-fitted estimator, or the present uniform studentized coverage theorem for admissible $\sigma_n \rightarrow 0$ collections. Their covariate-aware tightening paper supplies conditional-transport identification and bound-tightening structure, not observational semiparametric efficiency or Wald theory. Their direct conditional-optimal-transport estimation paper supplies adapted-Wasserstein continuity and an observational finite-sample value-rate route using cross-fitting and self-normalized propensity weighting; it does not derive the efficient influence function, efficiency attainment, studentizer consistency, or uniform Wald coverage. Ji–Lei–Spector supply observational cross-fitted AIPW inference for fitted dual transport bounds and the 401(k) SIPP application; that construction can retain weak-duality slack and does not establish efficiency for the sharp smooth primal endpoint or vanishing-variance

studentization. Manole–Balakrishnan–Niles–Weed–Wasserman supply the smooth unconditional Brenier stability, plug-in limit theory, Wald inference away from equality, and marginal-sample efficiency substrate; their model has neither the causal covariate-composition term nor observational inverse-propensity correction. Tan–Blanchet–Syrkanis study optimal-transport partial identification of policy-relevant treatment effects with instrumental variables, a different estimand and identifying design, and do not provide the present conditional observational efficiency theorem. Chernozhukov and coauthors and Newey–Robins provide the general orthogonal-score and cross-fitting templates; here the transport-specific ingredients are the signed quadratic Brenier remainder and the variance expansion that retains the shrinking canonical-gradient scale. At exact equality, the weighted-Neumann second variation and non-Hilbert–Schmidt full-tangent obstruction leave regularized or alternative adaptive calibration unresolved; the Gaussian-series existence and uniform-tail criteria themselves are settled.

3 Setup

Target (estimand / structural parameter): $\theta_n = E_{P_n}[W_2^2(Q_{0,P_n}(\cdot | X_n), Q_{1,P_n}(\cdot | X_n))]$. Identifying functional / recovery map: $\psi_n(O) = V_n(X) - \theta_n + \sum_{a \in \mathcal{A}} 1\{A = a\} \phi_{a,n}(X, Y) / \pi_{a,n}(X)$.

- n — integer; $\mathbb{N}$; index

Definition. Triangular-array index.

- i — integer; $\{1, \dots, n\}$; index

Definition. Unit index.

- j — integer; $\{1, \dots, n\}$; index

Definition. Training-sample index.

- K — integer; $\{2, 3, \dots\}$; constant

Definition. Fixed number of cross-fitting folds.

- k — integer; $\{1, \dots, K\}$; index

Definition. Fold index.

- d_X — integer; $\mathbb{N}$; dimension

Definition. Covariate dimension.

- d_Y — integer; $\mathbb{N}$; dimension

Definition. Outcome dimension.

- D — integer; $\mathbb{N}$; dimension

Definition. Joint conditional-density dimension $D = d_X + d_Y$.

- $\mathcal{X}$ — set; $\mathbb{R}^{d_X}$; support

Definition. Compact covariate support.

- $\mathcal{Y}$ — set; $\mathbb{R}^{d_Y}$; support

Definition. Compact outcome support.

- x — argument; $\mathcal{X}$; argument

Definition. Covariate argument.

- y — argument; $\mathcal{Y}$; argument

Definition. Outcome argument.

- y_0 — argument; $\mathcal{Y}$; argument

Definition. First transport-coordinate argument.

- y_1 — argument; $\mathcal{Y}$; argument

Definition. Second transport-coordinate argument.

- z — argument; $\mathcal{Y}$; argument

Definition. Integration argument.

- B — set; $\mathcal{B}(\mathcal{Y})$; argument

Definition. Borel outcome event.

- $\mathcal{A}$ — set; $\{0, 1\}$; index set

Definition. Treatment-state set.

- a — integer; $\mathcal{A}$; index

Definition. Treatment-state argument.

- P_n — probability law; $\mathcal{P}(\mathcal{X} \times \mathcal{A} \times \mathcal{Y})$; law

Definition. Observed-data law.

- X_n — random vector; $\mathcal{X}$; observed variable

Definition. Observed covariate under P_n .

- A_n — random variable; $\mathcal{A}$; observed variable

Definition. Observed treatment under P_n .

- Y_n — random vector; $\mathcal{Y}$; observed variable

Definition. Observed outcome under P_n .

- $O_{i,n}$ — random vector; $\mathcal{X} \times \mathcal{A} \times \mathcal{Y}$; observation

Definition. $O_{i,n} = (X_{i,n}, A_{i,n}, Y_{i,n})$.

- $Y_n(a)$ — random vector; $\mathcal{Y}$; potential outcome

Definition. Potential outcome under treatment state a on a causal extension.

- $P_{X,n}$ — probability law; $\mathcal{P}(\mathcal{X})$; marginal

Definition. Covariate marginal of P_n .

- $\ell_{\mathcal{Y}}$ — measure; $\mathcal{M}(\mathcal{Y})$; reference measure

Definition. Lebesgue measure restricted to $\mathcal{Y}$.

- π_n — function; $\mathcal{X} \rightarrow (0, 1)$; propensity

Definition. $\pi_n(x) = P_n(A_n = 1 \mid X_n = x)$.

- $\pi_{a,n}$ — function; $\mathcal{X} \rightarrow (0, 1)$; arm propensity

Definition. $\pi_{1,n} = \pi_n$ and $\pi_{0,n} = 1 - \pi_n$.

- Q_{a,P_n} — probability kernel; $\mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$; observed arm kernel

Definition. $Q_{a,P_n}(B \mid x) = P_n(Y_n \in B \mid A_n = a, X_n = x)$.

- $q_{a,n}$ — function; $\mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$; observed arm density

Definition. $Q_{a,P_n}(dy \mid x) = q_{a,n}(x, y)d\ell_{\mathcal{Y}}(y)$.

- μ_{a,P_n} — probability kernel; $\mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$; causal arm kernel

Definition. $\mu_{a,P_n}(\cdot \mid x) = \mathcal{L}_{P_n}(Y_n(a) \mid X_n = x)$.

- $\nu_{a,n}$ — probability law; $\mathcal{P}(\mathcal{X} \times \mathcal{Y})$; observed arm law

Definition. $\nu_{a,n}(dx, dy) = P_{X,n}(dx)Q_{a,P_n}(dy \mid x)$.

- ν_0 — probability law; $\mathcal{P}(\mathcal{Y})$; argument

Definition. First generic transport marginal.

- ν_1 — probability law; $\mathcal{P}(\mathcal{Y})$; argument

Definition. Second generic transport marginal.

- γ — probability law; $\mathcal{P}(\mathcal{Y}^2)$; argument

Definition. Generic transport coupling.

- $\Gamma(\nu_0, \nu_1)$ — set; $\mathcal{P}(\mathcal{Y}^2)$; coupling set

Definition. Set of couplings with marginals ν_0 and ν_1 .

- W_2^2 — functional; $\mathcal{P}(\mathcal{Y})^2 \rightarrow \mathbb{R}_+$; transport functional

Definition. $W_2^2(\nu_0, \nu_1) = \inf_{\gamma \in \Gamma(\nu_0, \nu_1)} \int \|y_1 - y_0\|_2^2 d\gamma(y_0, y_1)$.

- V_n — function; $\mathcal{X} \rightarrow \mathbb{R}_+$; conditional transport value

Definition. $V_n(x) = W_2^2(Q_{0,P_n}(\cdot \mid x), Q_{1,P_n}(\cdot \mid x))$.

- θ_n — scalar; $\mathbb{R}_+$; observed-law estimand

Definition. $\theta_n = E_{P_n}[V_n(X_n)]$.

- $u_{a,n}$ — function; $\mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$; dual potential

Definition. An optimal conditional quadratic Kantorovich dual potential for Q_{a,P_n} .

- $\phi_{a,n}$ — function; $\mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$; centered transport derivative

Definition. $\phi_{a,n}(x, y) = u_{a,n}(x, y) - \int u_{a,n}(x, z)Q_{a,P_n}(dz \mid x)$.

- ψ_n — function; $\mathcal{X} \times \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$; canonical gradient

Definition. $\psi_n(O) = V_n(X) - \theta_n + \sum_{a \in \mathcal{A}} 1\{A = a\}\phi_{a,n}(X, Y)/\pi_{a,n}(X)$.

- σ_n^2 — scalar; $\mathbb{R}_+$; efficiency bound

Definition. $\sigma_n^2 = E_{P_n}[\psi_n(O_{1,n})^2]$.

- σ_n — scalar; $\mathbb{R}_+$; standard deviation

Definition. $\sigma_n = (\sigma_n^2)^{1/2}$.

- ε — constant; $(0, 1/2)$; constant

Definition. Fixed overlap constant.

- c — constant; $(0, \infty)$; constant

Definition. Lower density bound.

- C — constant; $(0, \infty)$; constant

Definition. Upper density bound.

- s — constant; $(0, \infty)$; constant

Definition. Hölder smoothness index.

- L — constant; $(0, \infty)$; constant

Definition. Hölder radius.

- J_n — integer sequence; $\mathbb{N}$; tuning sequence

Definition. Wavelet or sieve resolution satisfying $2^{J_n D} \asymp n^{D/(2s+D)}$.

- β — constant; $(0, \infty)$; rate exponent

Definition. Clipped-propensity learning exponent.

- σ_* — constant; $(0, \infty)$; constant

Definition. Variance-separation constant.

- $I_{k,n}$ — set; $\mathcal{P}(\{1, \dots, n\})$; cross-fitting fold

Definition. Evaluation fold k .

- $\mathcal{T}_{-k,n}$ — sample; $(\mathcal{X} \times \mathcal{A} \times \mathcal{Y})^{n-|I_{k,n}|}$; training sample

Definition. Training complement $\{O_{j,n} : j \notin I_{k,n}\}$.

- $\hat{\pi}_{-k,n}$ — function; $\mathcal{X} \rightarrow (0, 1)$; learner

Definition. Propensity learner fitted on $\mathcal{T}_{-k,n}$.

- $\bar{\pi}_{-k,n}$ — function; $\mathcal{X} \rightarrow [\varepsilon, 1 - \varepsilon]$; learner

Definition. Clipped fitted propensity.

- $\bar{\pi}_{a,-k,n}$ — function; $\mathcal{X} \rightarrow [\varepsilon, 1 - \varepsilon]$; arm learner

Definition. $\bar{\pi}_{1,-k,n} = \bar{\pi}_{-k,n}$ and $\bar{\pi}_{0,-k,n} = 1 - \bar{\pi}_{-k,n}$.

- $\hat{q}_{a,-k,n}$ — jointly measurable density; $\mathcal{X} \times \mathcal{Y} \rightarrow [c, C]$; learner

Definition. Normalized fitted arm density satisfying $\int_{\mathcal{Y}} \hat{q}_{a,-k,n}(x, y) d\ell_{\mathcal{Y}}(y) = 1$ for almost every x .

- $\hat{Q}_{a,-k,n}$ — probability kernel; $\mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$; learner

Definition. Kernel induced by $\hat{q}_{a,-k,n}$.

- $\hat{u}_{a,-k,n}$ — function; $\mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$; learner

Definition. Fitted conditional quadratic dual potential.

- $\hat{\phi}_{a,-k,n}$ — function; $\mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$; learner

Definition. Centered fitted dual potential.

- $\hat{V}_{-k,n}$ — function; $\mathcal{X} \rightarrow \mathbb{R}_+$; learner

Definition. Plug-in conditional transport value from the fitted arm kernels.

- $\hat{\gamma}_{-k,n}$ — probability kernel; $\mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y}^2)$; learner

Definition. Optimal fitted conditional coupling.

- $\hat{S}_{-k,n}$ — function; $\mathcal{X} \times \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$; score learner

Definition. Held-out coupled one-step score.

- $r_{\pi,n}$ — positive sequence; $\mathbb{R}_+$; rate

Definition. Clipped-propensity learning rate.

- $r_{q,n}$ — positive sequence; $\mathbb{R}_+$; rate

Definition. Arm-density learning rate.

- $r_{\phi,n}$ — positive sequence; $\mathbb{R}_+$; derived rate

Definition. Centered-potential rate derived from $r_{q,n}$ by the proposed transport-stability argument.

- $r_{V,n}$ — positive sequence; $\mathbb{R}_+$; derived rate

Definition. Conditional-transport-value rate derived from $r_{q,n}$ by the proposed transport-stability argument.

- $r_{S,n}$ — positive sequence; $\mathbb{R}_+$; rate

Definition. Derived coupled-score learning scale.

- $r_{\sigma,n}$ — positive sequence; $\mathbb{R}_+$; rate

Definition. Derived empirical-variance learning scale.

- $\mathcal{M}_n^{\text{reg}}$ — set of laws; $\mathcal{P}(\mathcal{X} \times \mathcal{A} \times \mathcal{Y})$; class

Definition. Regular observed-law class.

- $\hat{\theta}_n$ — scalar; $\mathbb{R}$; estimator

Definition. Cross-fitted one-step estimator.

- $\hat{\sigma}_n^2$ — scalar; $\mathbb{R}_+$; variance learner

Definition. Out-of-fold empirical score variance.

- $\hat{\sigma}_n$ — scalar; $\mathbb{R}_+$; studentizer

Definition. Positive-root studentizer.

- $\mathcal{B}_{k,n}$ — scalar; $\mathbb{R}$; remainder term

Definition. Conditional foldwise score bias.

- R_n — scalar; $\mathbb{R}$; remainder term

Definition. Estimator remainder.

- α — scalar; $(0,1)$; inference constant

Definition. Nominal error probability.

- $z_{1-\alpha/2}$ — scalar; $\mathbb{R}$; inference object

Definition. Standard-normal critical value.

- Φ — function; $\mathbb{R} \rightarrow (0,1)$; reference distribution

Definition. Standard-normal distribution function.

- t_π — constant; $(0,\infty)$; constant

Definition. Hölder smoothness index for the propensity class.

- L_π — constant; $(0,\infty)$; constant

Definition. Hölder radius for the propensity class.

- λ — constant; $(0,\infty)$; constant

Definition. Strong-convexity modulus for the conditional Brenier potentials.

4 Assumptions

Assumption 1 (ass:observed-iid). $O_{1,n}, \dots, O_{n,n} \stackrel{\text{iid}}{\sim} P_n$.

Standard (*i.i.d. sampling*, [7]).

Assumption 2 (ass:consistency). $Y_n = A_n Y_n(1) + (1 - A_n) Y_n(0)$.

Standard (*consistency*, [9]).

Assumption 3 (ass:ignorability). $(Y_n(0), Y_n(1)) \perp A_n \mid X_n$.

Standard (*conditional ignorability*, [9]).

Assumption 4 (ass:fixed-overlap). $\varepsilon \leq \pi_n(X_n) \leq 1 - \varepsilon$ almost surely.

Standard (*strong overlap*, [10]).

Assumption 5 (ass:compact-support). $P_n(X_n \in \mathcal{X}, Y_n \in \mathcal{Y}) = 1$.

Standard (*compact support*, [1]).

Assumption 6 (ass:bounded-arm-densities). $c \leq q_{a,n}(x, y) \leq C$ for every $a \in \mathcal{A}$ and every $(x, y) \in \mathcal{X} \times \mathcal{Y}$.

Standard (*bounded conditional densities*, [1]).

Assumption 7 (ass:holder-arm-densities). $q_{a,n} \in \Sigma(s, L; \mathcal{X} \times \mathcal{Y})$ for every $a \in \mathcal{A}$.

Standard (*Hölder-ball conditional densities*, [1]).

Assumption 8 (ass:variance-separated). $\sigma_n \geq \sigma_*$.

Standard (*uniform nondegenerate variance*, [7]).

Assumption 9 (ass:regular-support-geometry). $\mathcal{X}$ is the closure of a bounded connected Lipschitz domain in $\mathbb{R}^{d_x}$, and $\mathcal{Y}$ is the closure of a bounded connected uniformly convex C^2 domain with nonempty interior in $\mathbb{R}^{d_y}$.

Novel. standard: connected-support first variation in Manole et al. (2024), Lemma 23, and standard global Brenier regularity on uniformly convex domains

Assumption 10 (ass:uniform-conditional-brenier-smoothness). With $\mathfrak{F}$ denoting the feasible conditional-density class in *def : coupled – fold – learners*, the quadratic Brenier potential $b_{f_0, f_1, x}$ transporting $f_0(x, \cdot)$ to $f_1(x, \cdot)$ belongs to $C^2(\mathcal{Y})$ for every $f_0, f_1 \in \mathfrak{F}$ and almost every x .

Standard (*uniform A1-type Brenier smoothness*, manole2024plugin, [6]).

Assumption 11 (ass:uniform-conditional-brenier-lower-curvature). There exists a fixed constant $m_{OT} > 0$ such that $\nabla_y^2 b_{f_0, f_1, x}(y) \succeq m_{OT} I_{d_y}$ for every $f_0, f_1 \in \mathfrak{F}$, almost every x , and every $y \in \mathcal{Y}$.

Standard (*uniform A1-type lower Brenier curvature*, manole2024plugin, [6]).

Assumption 12 (ass:uniform-conditional-brenier-upper-curvature). There exists a fixed constant $M_{OT} < \infty$ such that $\nabla_y^2 b_{f_0, f_1, x}(y) \preceq M_{OT} I_{d_y}$ for every $f_0, f_1 \in \mathfrak{F}$, almost every x , and every $y \in \mathcal{Y}$.

Standard (*uniform A1-type upper Brenier curvature*, manole2024plugin, [6]).

Assumption 13 (ass:holder-propensity). There exist fixed constants $t_\pi, L_\pi > 0$ such that $\pi_n \in \Sigma(t_\pi, L_\pi; \mathcal{X})$.

Standard (*Hölder nonparametric Bernoulli regression*, lin2026statistical, [1]).

5 Definitions

Definition 1 (def:regular-model-class). Defined object. *Smooth observed-law parent class*

Construction.

$\mathcal{M}_n^{\text{reg}} = \{P_n \in \mathcal{P}(\mathcal{X} \times \mathcal{A} \times \mathcal{Y}) : P_n \text{ satisfies } \text{ass} : \text{fixed-overlap}, \text{ass} : \text{compact-support}, \text{ass} : \text{bounded-arm-densities}, \text{ass} : \text{holder-arm-densities}\}$
. No lower bound is imposed on σ_n^2 ; in particular, $\sigma_n^2 = 0$ is permitted whenever the displayed member properties hold.

Member properties. *ass:fixed-overlap; ass:compact-support; ass:bounded-arm-densities; ass:holder-arm-densities.*

Definition 2 (def:coupled-fold-learners). Defined object. *Normalized coupled foldwise transport learners*
Construction. Choose deterministic folds satisfying $\max_{1 \leq k \leq K} \|I_{k,n}\| - n/K \leq 1$. Let $\mathfrak{F}$ be the class of $f \in \Sigma(s, L; \mathcal{X} \times \mathcal{Y})$ such that $c \leq f \leq C$ and $\int_{\mathcal{Y}} f(x, y) d\ell_{\mathcal{Y}}(y) = 1$ for almost every x . Let $\mathfrak{F}_{J_n}$ be a fixed finite L^∞ -net of $\mathfrak{F}$, constructed by quantizing boundary-adapted tensor-wavelet coefficients through level J_n , with mesh order $2^{-J_n s}$ and logarithmic cardinality order $2^{J_n D}$. Put $N_{a,-k,n} = \sum_{j \notin I_{k,n}} 1\{A_{j,n} = a\}$. If $N_{a,-k,n} > 0$, define $\hat{q}_{a,-k,n}$ as the lexicographically first minimizer over $f \in \mathfrak{F}_{J_n}$ of

$$\frac{1}{N_{a,-k,n}} \sum_{j \notin I_{k,n} : A_{j,n} = a} \left\{ \int_{\mathcal{Y}} f(X_{j,n}, y)^2 d\ell_{\mathcal{Y}}(y) - 2f(X_{j,n}, Y_{j,n}) \right\}.$$

If $N_{a,-k,n} = 0$, use the first element of $\mathfrak{F}_{J_n}$. For every k, n , fix before observing the data a rule $\mathbf{A}_{\pi,k,n}$ such that $(t, x) \mapsto \mathbf{A}_{\pi,k,n}(t)(x) \in (0, 1)$ is jointly measurable for a training sample t and $x \in \mathcal{X}$. Set $\hat{\pi}_{-k,n} = \mathbf{A}_{\pi,k,n}(\mathcal{T}_{-k,n})$, set $\bar{\pi}_{-k,n} = \min\{1 - \varepsilon, \max\{\varepsilon, \hat{\pi}_{-k,n}\}\}$, and define $\hat{Q}_{a,-k,n}(B | x) = \int_B \hat{q}_{a,-k,n}(x, y) d\ell_{\mathcal{Y}}(y)$. Select the conditional quadratic optimizer and a mutually c -conjugate dual pair measurably by the minimum-norm normalized tie-break, obtaining $x \mapsto \hat{\gamma}_{-k,n}(\cdot | x)$ and $x \mapsto (\hat{u}_{0,-k,n}, \hat{u}_{1,-k,n})$. Require dual feasibility, contact on $\hat{\gamma}_{-k,n}$, and $\hat{V}_{-k,n}(x) = \sum_{a \in \mathcal{A}} \int \hat{u}_{a,-k,n}(x, y) \hat{Q}_{a,-k,n}(dy | x)$. Finally set $\hat{\phi}_{a,-k,n} = \hat{u}_{a,-k,n} - \int \hat{u}_{a,-k,n} d\hat{Q}_{a,-k,n}$, so its fitted-kernel mean is exactly zero.

Inputs. $\mathcal{T}_{-k,n}; \hat{\pi}_{-k,n}; \hat{q}_{a,-k,n}; \varepsilon; \ell_{\mathcal{Y}}$.

Definition 3 (def:foldwise-rate-route). Defined object. *Foldwise Hölder and propensity rate route*

Construction. Use the finite-net conditional-density empirical-risk minimizer in def: coupled-fold-learners with $2^{J_n D} \asymp n^{D/(2s+D)}$. Define $r_{q,n} = n^{-s/(2s+D)}$, $r_{\phi,n} = r_{q,n}$, $r_{V,n} = r_{q,n}$, $r_{S,n} = r_{\pi,n} + 2r_{q,n}$, and $r_{\sigma,n} = r_{S,n} + n^{-1/2}$. The density rate is derived from approximation error $2^{-J_n s}$, entropy $\log |\mathfrak{F}_{J_n}| \lesssim 2^{J_n D}$, and the empirical conditional-density loss. The transport-potential, value, signed quadratic-remainder, score, and variance rates are conclusions derived from the three all-feasible-pairs Brenier assumptions, dual contact, exact centering, and cross-fitting. No propensity smoothness or propensity rate is part of this density route.

Inputs. $J_n; D; s; \hat{q}_{a,-k,n}$.

Definition 4 (def:held-out-score). Defined object. *Held-out coupled one-step score*

Construction.

$\hat{S}_{-k,n}(O_{i,n}) = \hat{V}_{-k,n}(X_{i,n}) + A_{i,n} \hat{\phi}_{1,-k,n}(X_{i,n}, Y_{i,n}) / \bar{\pi}_{-k,n}(X_{i,n}) + (1 - A_{i,n}) \hat{\phi}_{0,-k,n}(X_{i,n}, Y_{i,n}) / [1 - \bar{\pi}_{-k,n}(X_{i,n})]$, for $i \in I_{k,n}$.

Inputs. $O_{i,n}; \hat{V}_{-k,n}; \hat{\phi}_{a,-k,n}; \bar{\pi}_{-k,n}$.

Definition 5 (def:one-step-estimator). Defined object. *Cross-fitted one-step estimator*

Construction. $\hat{\theta}_n = n^{-1} \sum_{k=1}^K \sum_{i \in I_{k,n}} \hat{S}_{-k,n}(O_{i,n})$.

Inputs. $I_{k,n}; \hat{S}_{-k,n}$.

Definition 6 (def:empirical-studentizer). Defined object. *Empirical score studentizer*

Construction.

$$\hat{\sigma}_n^2 = n^{-1} \sum_{k=1}^K \sum_{i \in I_{k,n}} [\hat{S}_{-k,n}(O_{i,n}) - \hat{\theta}_n]^2$$

and $\hat{\sigma}_n = (\hat{\sigma}_n^2)^{1/2}$. In every studentized statistic in this core, define a quotient with denominator $\hat{\sigma}_n = 0$ to equal zero.

Inputs. $I_{k,n}; \hat{S}_{-k,n}; \hat{\theta}_n$.

Definition 7 (def:degeneracy-handle). Defined object. *Second-order conditional-transport handle*

Construction. *Derive second conditional Brenier derivatives along local perturbations of Q_{0,P_n} and Q_{1,P_n} , then combine their observed-data scores with a derivative-aware resampling rule.*

Inputs. $Q_{a,P_n}; \phi_{a,n}; \sigma_n$.

Definition 8 (def:predetermined-triangular-law-class). Defined object. *Predetermined variance-unrestricted triangular parent*

Construction.

$\mathcal{P}^{\text{vu}} = \mathcal{P} = \{\mathbf{P} = (P_n)_{n \geq 1} : \text{for every } n, O_{1,n}, \dots, O_{n,n} \stackrel{\text{iid}}{\sim} P_n, P_n \in \mathcal{M}_n^{\text{reg}}, \text{ and } \text{ass:regular-support-geometry, ass:uniform-conditional-brenier-smoothness, ass:uniform-conditional-brenier-lower-curvature, and ass:uniform-conditional-brenier-upper-curvature hold}\}$. The class is fixed before observing data. It contains no learner-rate, remainder, coverage, efficiency, or other estimator-side conclusion, and it explicitly permits $\sigma_n = 0$. The notation $\mathcal{P}$ is retained as a backward-compatible alias for $\mathcal{P}^{\text{vu}}$. Uniformity at row n means $\sup_{\mathbf{P} \in \mathcal{P}^{\text{vu}}} P_n(\cdot)$.

Member properties. *ass:observed-iid; ass:fixed-overlap; ass:compact-support; ass:bounded-arm-densities; ass:holder-arm-densities; ass:regular-support-geometry; ass:uniform-conditional-brenier-smoothness; ass:uniform-conditional-brenier-lower-curvature; ass:uniform-conditional-brenier-upper-curvature.*

Definition 9 (def:propensity-rate-regimes). Defined object. *Generic and primitive propensity-rate regimes*

Construction. Let $\mathcal{C} \subseteq \mathcal{P}^{\text{vu}}$ be any predetermined collection. Fix before observing data a jointly measurable learner rule on every training complement and a deterministic sequence $r_{\pi,n} \rightarrow 0$. The generic estimator-side learner premise over $\mathcal{C}$ is

$$\lim_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{\mathbf{P} \in \mathcal{C}} P_n \left(\max_k \|\bar{\pi}_{-k,n} - \pi_n\|_{L^2(P_{X,n})} > M r_{\pi,n} \right) = 0.$$

This premise does not define membership in $\mathcal{C}$, $\mathcal{P}^{\text{vu}}$, or $\mathcal{P}_{\sigma_*}^{\text{vs}}$. For the primitive Hölder regime, work on $\mathcal{P}_{\sigma_*, t_\pi, L_\pi}^{\text{vs,H}}$. Let $\mathfrak{G}_J$ be a deterministic finite L^∞ -net of $\{g \in \Sigma(t_\pi, L_\pi; \mathcal{X}) : \varepsilon \leq g \leq 1 - \varepsilon\}$, with mesh $2^{-J t_\pi}$ and $\log |\mathfrak{G}_J| \lesssim 2^{J d_X}$. On each training complement choose the lexicographically first minimizer of $\sum_j (A_{j,n} - g(X_{j,n}))^2$ over $g \in \mathfrak{G}_J$, clip it to $[\varepsilon, 1 - \varepsilon]$, and choose $2^{J d_X} \asymp n^{d_X / (2t_\pi + d_X)}$.

Inputs. $\mathcal{T}_{-k,n}; \bar{\pi}_{-k,n}; r_{\pi,n}; \varepsilon$.

Definition 10 (def:variance-separated-triangular-law-class). Defined object. *Variance-separated triangular inference class*

Construction. For a fixed $\sigma_* > 0$,

$$\mathcal{P}_{\sigma_*}^{\text{vs}} = \left\{ \mathbf{P} \in \mathcal{P}^{\text{vu}} : \inf_{n \geq 1} \sigma_n \geq \sigma_* \right\}.$$

This is a structural subclass of the variance-unrestricted parent. It contains no learner-rate, remainder, coverage, or efficiency conclusion.

Member properties. *ass:variance-separated.*

Definition 11 (def:holder-propensity-triangular-law-class). Defined object. *Variance-separated H"older-propensity subclass*

Construction. For fixed $t_\pi, L_\pi > 0$,

$$\mathcal{P}_{\sigma_*, t_\pi, L_\pi}^{\text{vs,H}} = \left\{ \mathbf{P} \in \mathcal{P}_{\sigma_*}^{\text{vs}} : \pi_n \in \Sigma(t_\pi, L_\pi; \mathcal{X}) \text{ for every } n \right\}.$$

This class adds only the structural smoothness of each row propensity; it does not include performance of any fitted learner.

Member properties. *ass:holder-propensity.*

Definition 12 (def:unique-normalized-conditional-potentials). Defined object. *Unique normalized conditional transport potentials*

Construction. For an observed law P_n , write $U(P_n)$ when, for $P_{X,n}$ -almost every x , a mutually conjugate optimal quadratic dual pair for $Q_{0,P_n}(\cdot | x)$ and $Q_{1,P_n}(\cdot | x)$ exists, admits a jointly measurable version in (x, y) , and is unique after imposing $\min_{y \in \mathcal{Y}} u_{0,n}(x, y) = 0$.

Inputs. $P_n; Q_{a,P_n}; u_{a,n}$.

Definition 13 (def:conditional-transport-pathwise-differentiability). Defined object. *Conditional transport pathwise differentiability property*

Construction. For an observed law P_n , write $D(P_n)$ when every regular dominated one-dimensional observed-data submodel through P_n , with score $h \in L_0^2(P_n)$, satisfies

$$\dot{\theta}_n(h) = E[(V_n(X_n) - \theta_n)h_X(X_n)] + \sum_{a \in \mathcal{A}} E_{P_{X,n}} \left[\int \phi_{a,n}(X_n, y) h_{Y,a}(X_n, y) Q_{a,P_n}(dy | X_n) \right],$$

where $h_X(x) = E[h(O) | X = x]$ and $h_{Y,a}(x, y) = h(x, a, y) - E[h(O) | A = a, X = x]$.

Inputs. $P_n; \theta_n; V_n; \phi_{a,n}$.

Definition 14 (def:gradient-square-integrability). Defined object. *Observed-gradient square-integrability property*

Construction. For an observed law P_n , write $G(P_n)$ when the displayed observed-data gradient satisfies $\psi_n \in L_0^2(P_n)$.

Inputs. $P_n; \psi_n$.

6 Main results

Open-ended Question 1 (oeq:degeneracy-adaptive-inference). *The Gaussian spectral subproblem is settled: square summability is necessary and sufficient for L^2 existence of the centered Gaussian quadratic series, and a uniform square-summable tail is necessary and sufficient for uniform finite-rank approximation. Does there exist an honest, degeneracy-adaptive confidence procedure uniformly over the full variance-unrestricted parent, outside the ruled-out unregularized full-tangent Gaussian spectral route, that is first-order efficient away from equality and correctly calibrated at equality; and which covariate-direction regularization, regularized-Hessian estimability, or alternative derivative-aware calibration conditions are necessary and sufficient?*

The exact frontier currently proved is as follows. At every fixed positive-variance regular law, ψ_n is the canonical gradient and σ_n^2 is the efficiency bound. On positive-variance triangular collections satisfying

$$\frac{\sqrt{n} \delta_n}{\sigma_n} \rightarrow 0, \quad \frac{r_{S,n}}{\sigma_n} \rightarrow 0, \quad \sqrt{n} \sigma_n \rightarrow \infty$$

uniformly, the cross-fitted estimator is asymptotically linear with influence function ψ_n , its studentizer is ratio consistent, and its Wald interval has uniform asymptotic coverage. At conditional equality, $\theta_n = 0$, $\psi_n = 0$, and purely first-order Gaussian calibration fails. For a centered Gaussian quadratic calibration, square summability is necessary and sufficient for L^2 existence, and a uniformly vanishing square-summable tail is necessary and sufficient for uniform finite-rank approximation. The full parent violates these requirements at the identity-compression equality witness. Consequently, removing that compression, obtaining a uniform Hilbert–Schmidt tail, and uniformly estimating the regularized Hessian are necessary for the regularized spectral route; their primitive sufficiency, and the existence or impossibility of an alternative derivative-aware route, remain unresolved.

Justification. The exact spectral obstruction identifies what a successful adaptive procedure must overcome, but the frozen parent supplies no uniformly estimable regularization or alternative calibration. *Gap.* Necessary and sufficient primitive conditions for full-parent honest adaptation remain unknown. *Consumer.* A solution would determine uncertainty reporting when the conditional outcome laws are nearly or exactly equal.

Lemma 1 (lem:manole-unconditional-smooth-wasserstein-efficiency). *For the unconditional functional $W_2^2(P, Q)$ based on one sample from one marginal or two independent marginal samples, Manole, Balakrishnan, Niles-Weed, and Wasserman derive centered plug-in central limit theorems, consistent plug-in variance estimators and Wald intervals away from $P = Q$, the efficient one-sample influence function given by the centered Kantorovich potential, and one- and two-sample local asymptotic minimax lower bounds matching the corresponding potential variances under their smooth positive-density conditions.*

Lemma 2 (lem:canonical-gradient-information-inequality). *In a dominated semiparametric model, if a scalar parameter has canonical gradient $\psi \in L_0^2(P)$, then every regular asymptotically linear estimator has asymptotic variance at least $E_P[\psi^2]$. An asymptotically linear estimator whose influence function is ψ attains this lower bound.*

Lemma 3 (lem:finite-net-conditional-density-rate). *For the constrained finite-net empirical-risk learners in def : coupled – fold – learners, under i.i.d. sampling, fixed overlap, bounded arm densities, and Hölder arm densities,*

$$\max_{k,a} \|\widehat{q}_{a,-k,n} - q_{a,n}\|_{L^2(P_{X,n} \otimes \ell_Y)} = O_{P_n}(2^{-J_n s} + \sqrt{2^{J_n D}/n}) = O_{P_n}(n^{-s/(2s+D)})$$

uniformly over $\mathbf{P} \in \mathcal{P}$.

Proof. Fix a fold and arm and condition on the training treatment labels. If N is the arm count and $m \asymp n$ the training size, overlap gives $P(N < \varepsilon m/2) \leq e^{-c_0 m}$ uniformly. Conditional on N , the selected observations are independent with law $P_n(\cdot | A = a)$. For

$$\ell_f(x, y) = \int f(x, z)^2 d\ell_Y(z) - 2f(x, y),$$

conditioning on X yields the exact excess-risk identity

$$\mathcal{R}(f) = E[\ell_f - \ell_q | A = a] = \|f - q\|_{L^2(P_{X|A=a} \otimes \ell_Y)}^2. \quad (1)$$

Uniform boundedness gives $|\ell_f - \ell_q| \leq B$ and a Bernstein variance bound $\text{Var}(\ell_f - \ell_q | A = a) \leq B\mathcal{R}(f)$. Bernstein's inequality and a union bound over $\mathfrak{F}_{J_n}$ therefore imply, with probability at least $1 - 2e^{-u}$, simultaneously for every candidate,

$$|(\mathbb{P}_N - P)(\ell_f - \ell_q)| \leq \frac{1}{4}\mathcal{R}(f) + C_1 \frac{\log |\mathfrak{F}_{J_n}| + u}{N}. \quad (2)$$

By the net construction there exists f° with $\|f^\circ - q\|_\infty \leq C_2 2^{-J_n s}$, while $\log |\mathfrak{F}_{J_n}| \lesssim 2^{J_n D}$. The empirical minimizing property, applied to (2) for $\widehat{q}$ and f° , gives

$$\mathcal{R}(\widehat{q}) \leq C_3 \left\{ 2^{-2J_n s} + \frac{2^{J_n D} + u}{n} \right\}. \quad (3)$$

Overlap makes the $L^2(P_{X|A=a} \otimes \ell_Y)$ and $L^2(P_X \otimes \ell_Y)$ norms uniformly equivalent. Taking square roots, a union bound over fixed $2K$ fold–arm pairs, and $2^{J_n D} \asymp n^{D/(2s+D)}$ yield

$$\max_{k,a} \|\widehat{q}_{a,-k,n} - q_{a,n}\|_2 = O_{P_n} \left(2^{-J_n s} + \sqrt{2^{J_n D}/n} \right) = O_{P_n} \left(n^{-s/(2s+D)} \right)$$

uniformly over $\mathcal{P}$. □

Lemma 4 (lem:conditional-coupling-sharpness). *Let P_X be a probability law on $\mathcal{X}$, and let Q_0, Q_1 be jointly measurable probability kernels from $\mathcal{X}$ to the compact metric space $\mathcal{Y}$. Among all probability laws $\tilde{P}$ on $\mathcal{X} \times \mathcal{Y}^2$ whose X -marginal is P_X and whose conditional Y_a -marginal given $X = x$ is $Q_a(\cdot | x)$,*

$$\inf_{\tilde{P}} E_{\tilde{P}}[\|Y_1 - Y_0\|_2^2] = \int_{\mathcal{X}} W_2^2(Q_0(\cdot | x), Q_1(\cdot | x)) dP_X(x).$$

The infimum is attained by a jointly measurable conditional optimal-coupling kernel. If, in addition, $A | X = x$ is drawn independently of $(Y_0, Y_1) | X = x$ with probabilities $\pi_a(x) > 0$, and $Y = Y_A$, the resulting law is a causal extension satisfying consistency and conditional ignorability and having observed arm kernels Q_a . Hence the displayed value is a sharp lower endpoint over causal extensions with these conditional marginals.

Proof. Disintegrate any admissible law as $P_X(dx)\gamma_x(dy_0, dy_1)$. The marginal restrictions imply $\gamma_x \in \Gamma(Q_0(\cdot | x), Q_1(\cdot | x))$ almost surely, so pointwise optimality and Tonelli give

$$E\|Y_1 - Y_0\|_2^2 \geq \int W_2^2(Q_0(\cdot | x), Q_1(\cdot | x))dP_X(x). \quad (1)$$

For attainment, $\mathcal{P}(\mathcal{Y}^2)$ is compact in the weak topology because $\mathcal{Y}$ is compact. For each x , the coupling constraint is a nonempty compact section and the continuous quadratic objective has a nonempty compact argmin section. Its graph is measurable: marginal equalities may be tested on a fixed countable dense subset of $C(\mathcal{Y})$, and the optimal value is the limit of finite-partition transport programs with measurable coefficients. A measurable selector can be constructed by successively intersecting each section with the first ball, from a countable dense family, of radius 2^{-m} that meets the preceding section. The nested sections have diameter tending to zero, and their limit γ_x^* is measurable and optimal. Thus $P_X(dx)\gamma_x^*(dy_0, dy_1)$ attains equality in (1).

Finally draw $A | X = x$ independently of $(Y_0, Y_1) | X = x$, with probabilities $\pi_a(x) > 0$, and set $Y = Y_A$. Then consistency and conditional ignorability hold and $\mathcal{L}(Y | A = a, X = x) = Q_a(\cdot | x)$. Hence the minimizing coupling is a compatible causal extension, proving sharpness. $\square$

Lemma 5 (lem:uniform-canonical-gradient-bound). *There is a finite constant M_ψ , depending only on ε and $\text{diam}(\mathcal{Y})$, such that every $P_n \in \mathcal{M}_n^{\text{reg}}$ satisfying ass : regular – support – geometry obeys $E_{P_n}[\psi_n(O_{1,n})] = 0$ and $|\psi_n(O_{1,n})| \leq M_\psi$ almost surely. Consequently, $\sup_{n, P_n} E_{P_n}[|\psi_n|^r] \leq M_\psi^r$ for every $r \geq 1$.*

Proof. Let $R = \text{diam}(\mathcal{Y})$. Since the quadratic cost lies in $[0, R^2]$, $0 \leq V_n(x), \theta_n \leq R^2$. For $y, y', z \in \mathcal{Y}$,

$$|\|y - z\|_2^2 - \|y' - z\|_2^2| \leq 2R\|y - y'\|_2 \leq 2R^2.$$

Taking the relevant conjugacy infimum or supremum preserves this oscillation bound, so $\text{osc}(u_{a,n}(x, \cdot)) \leq 2R^2$, and centering gives $|\phi_{a,n}| \leq 2R^2$. Exactly one arm indicator is nonzero and $\pi_{a,n} \geq \varepsilon$, hence

$$|\psi_n(O)| \leq R^2 + \frac{2R^2}{\varepsilon} =: M_\psi.$$

Moreover,

$$E\left[\frac{1\{A = a\}}{\pi_{a,n}(X)}\phi_{a,n}(X, Y) \middle| X\right] = \int \phi_{a,n}(X, y)Q_{a, P_n}(dy | X) = 0,$$

and $E[V_n(X) - \theta_n] = 0$. Thus $E\psi_n = 0$. The almost-sure envelope gives $E|\psi_n|^r \leq M_\psi^r$ for every $r \geq 1$, uniformly. $\square$

Lemma 6 (lem:wasserstein-l2-density-control). *If $\mathcal{Y}$ is a bounded connected Lipschitz domain and $f, g \in L^2(\mathcal{Y})$ are probability densities satisfying $f, g \geq c > 0$, then*

$$W_2^2(f d\ell_{\mathcal{Y}}, g d\ell_{\mathcal{Y}}) \leq c^{-1} C_{\mathcal{Y}}^2 \|f - g\|_{L^2(\ell_{\mathcal{Y}})}^2,$$

where $C_{\mathcal{Y}}$ is a Poincaré constant of $\mathcal{Y}$.

Proof. Put $h = g - f$, so $\int_{\mathcal{Y}} h d\ell_{\mathcal{Y}} = 0$. On the mean-zero subspace of $H^1(\mathcal{Y})$, the Poincaré inequality makes the functional

$$\eta \mapsto \frac{1}{2} \int \|\nabla \eta\|_2^2 d\ell_{\mathcal{Y}} - \int h \eta d\ell_{\mathcal{Y}}$$

coercive and strictly convex. Its minimizer φ satisfies

$$\int \nabla \varphi \cdot \nabla \eta = \int h \eta, \quad \|\nabla \varphi\|_2 \leq C_{\mathcal{Y}} \|h\|_2. \quad (1)$$

Set $\rho_t = (1 - t)f + tg$, $j_t = \nabla \varphi$, and $v_t = j_t / \rho_t$. Then $\partial_t \rho_t + \nabla \cdot j_t = 0$, the weak boundary flux is zero, and $\rho_t \geq c$. The dynamic transport bound obtained by regularizing this weak continuity equation and passing to the limit gives

$$W_2^2(f d\ell_{\mathcal{Y}}, g d\ell_{\mathcal{Y}}) \leq \int_0^1 \int \rho_t \|v_t\|_2^2 d\ell_{\mathcal{Y}} dt. \quad (2)$$

Using $\rho_t \geq c$ and (1) in (2) yields

$$W_2^2(f d\ell_Y, g d\ell_Y) \leq c^{-1} \|\nabla \varphi\|_2^2 \leq c^{-1} C_Y^2 \|f - g\|_2^2.$$

□

Lemma 7 (lem:equality-degeneracy-and-second-variation). *Let $P_n \in \mathcal{M}_n^{\text{reg}}$ satisfy **ass:regular-support-geometry**. Then $\sigma_n^2 = 0$ if and only if $Q_{0,P_n}(\cdot | x) = Q_{1,P_n}(\cdot | x)$ for $P_{X,n}$ -almost every x . At such a law, $\theta_n = 0$, $\psi_n = 0$ almost surely, and the zero-denominator oracle first-order pivot is identically zero and has Kolmogorov distance $1/2$ from $N(0,1)$. Moreover, suppress the row index and consider a path that fixes P_X and π , starts at conditional equality $q_{0,0} = q_{1,0} = q$, and satisfies $q_{a,t} = q + th_a$, where $\int_Y h_a(x, y) d\ell_Y(y) = 0$. Suppose the path remains in $\mathcal{M}_n^{\text{reg}}$, the functions q, h_0, h_1 are twice continuously differentiable in y with uniform bounds, and the mean-zero weighted Neumann solution w_x satisfies*

$$-\nabla_y \cdot \{q(x, y) \nabla_y w_x(y)\} = h_1(x, y) - h_0(x, y), \quad q(x, y) \nabla_y w_x(y) \cdot \mathbf{n}(y) = 0,$$

with w_x , $\nabla_y w_x$, and $\nabla_y^2 w_x$ jointly measurable and uniformly bounded. Then

$$\theta(P_t) = t^2 \mathcal{E} + o(t^2), \quad \mathcal{E} = \int_{\mathcal{X} \times \mathcal{Y}} q(x, y) \|\nabla_y w_x(y)\|_2^2 d(P_X \otimes \ell_Y)(x, y),$$

so the second directional derivative at equality is $2\mathcal{E}$.

Proof. Suppress n . Suppose first that $Q_0(\cdot | x) = Q_1(\cdot | x) = Q(\cdot | x)$. The diagonal coupling has zero cost, so $V(x) = 0$ and $\theta = 0$. Complementary slackness for any continuous mutually conjugate optimal pair gives $u_0(x, y) + u_1(x, y) = 0$ on the support of Q , hence everywhere because the density is positive and the potentials are continuous. Dual feasibility gives

$$|u_1(x, z) - u_1(x, y)| \leq \|z - y\|_2^2. \quad (1)$$

Partitioning the segment from y to z into m equal pieces makes the right side $\|z - y\|_2^2/m$; letting $m \rightarrow \infty$ shows that both potentials are constant. Thus $\phi_a = 0$, $\psi = 0$, and $\sigma^2 = 0$.

Conversely, if $\sigma^2 = 0$, then $\psi = 0$ almost surely. Conditional on $(X, A) = (x, a)$, fixed overlap gives

$$V(x) - \theta + \frac{\phi_a(x, Y)}{\pi_a(x)} = 0 \quad Q_a(\cdot | x)\text{-almost surely.} \quad (2)$$

Integrating (2) against Q_a gives $V(x) = \theta$, and substituting back gives $\phi_a(x, \cdot) = 0$. Positivity of q_a and continuity make each $u_a(x, \cdot)$ constant. Their sum is the dual value $V(x)$, while dual feasibility at $y_0 = y_1$ makes the sum nonpositive. Since $V(x) \geq 0$, one has $V(x) = 0$, and the metric property of W_2 gives $Q_0(\cdot | x) = Q_1(\cdot | x)$. Also $\theta = 0$. The frozen convention makes the oracle pivot identically zero, and therefore its Kolmogorov distance from $N(0,1)$ is $1/2$.

For the second variation put $\delta_x = h_1(x, \cdot) - h_0(x, \cdot)$,

$$\rho_{r,t} = q + t\{(1-r)h_0 + rh_1\}, \quad j_{r,t} = tq \nabla_y w_x, \quad 0 \leq r \leq 1.$$

The weighted Neumann equation gives $\partial_r \rho_{r,t} + \nabla_y \cdot j_{r,t} = 0$ and zero normal flux. Because $\rho_{r,t}$ is uniformly bounded below, the flow of $v_{r,t} = j_{r,t}/\rho_{r,t}$ couples the two endpoint densities. Cauchy–Schwarz along flow lines yields

$$W_2^2(q_{0,t} d\ell_Y, q_{1,t} d\ell_Y) \leq \int_0^1 \int \frac{\|j_{r,t}\|_2^2}{\rho_{r,t}} d\ell_Y dr = t^2 E_x + O(|t|^3), \quad (3)$$

where $E_x = \int q(x, y) \|\nabla_y w_x(y)\|_2^2 d\ell_Y(y)$, and the remainder is uniform in x . For the reverse bound let $\gamma_{x,t}$ be optimal and $W_{x,t}^2 = \int \|y_1 - y_0\|_2^2 d\gamma_{x,t}$. The marginal constraints and weak Neumann identity give

$$tE_x = \int \{w_x(y_1) - w_x(y_0)\} d\gamma_{x,t}. \quad (4)$$

Taylor expansion on convex $\mathcal{Y}$, followed by Cauchy–Schwarz, gives

$$|t|E_x \leq \{E_x + O(|t|)\}^{1/2} W_{x,t} + \frac{1}{2} \|\nabla_y^2 w_x\|_\infty W_{x,t}^2. \quad (5)$$

Equation (3) gives $W_{x,t} = O(|t|)$. If $E_x > 0$, divide (5) by $|t|$ and take a lower limit; if $E_x = 0$, positivity of q gives $\nabla w_x = 0$ and then $\delta_x = 0$. In either case, together with (3),

$$W_{x,t}^2 = t^2 E_x + o(t^2). \quad (6)$$

The uniform upper bound from (3) permits dominated convergence over P_X , so $\theta(P_t) = t^2 \int E_x dP_X(x) + o(t^2) = t^2 \mathcal{E} + o(t^2)$. Hence the second directional derivative is $2\mathcal{E}$. $\square$

Lemma 8 (lem:gaussian-quadratic-spectral-criterion). *Let $Z_1, Z_2, \dots$ be independent $N(0, 1)$ variables and let $\lambda_1, \lambda_2, \dots \in \mathbb{R}$. The centered quadratic series $\sum_{j \geq 1} \lambda_j (Z_j^2 - 1)$ converges in L^2 if and only if $\sum_{j \geq 1} \lambda_j^2 < \infty$. If $\lambda_j \geq 0$, the uncentered series $\sum_{j \geq 1} \lambda_j Z_j^2$ converges in L^1 if and only if $\sum_{j \geq 1} \lambda_j < \infty$. For a family Λ of square-summable sequences, common rank- J truncations approximate the centered series uniformly in L^2 if and only if $\lim_{J \rightarrow \infty} \sup_{\lambda \in \Lambda} \sum_{j > J} \lambda_j^2 = 0$. If $\lambda_j = \lambda_o > 0$ for infinitely many j , the centered partial sums have no weakly convergent subsequence.*

Proof. Let $S_J = \sum_{j=1}^J \lambda_j (Z_j^2 - 1)$. Independence, centering, and $E[(Z_j^2 - 1)^2] = 2$ give, for $J > L$,

$$E[(S_J - S_L)^2] = 2 \sum_{j=L+1}^J \lambda_j^2. \quad (1)$$

Thus (S_J) is Cauchy in L^2 exactly when $\sum_j \lambda_j^2 < \infty$. Completeness of L^2 proves the first assertion. Taking the supremum over $\lambda \in \Lambda$ in (1) proves both directions of the uniform truncation criterion.

If $\lambda_j \geq 0$, monotone convergence gives

$$E \left[\sum_{j \geq 1} \lambda_j Z_j^2 \right] = \sum_{j \geq 1} \lambda_j. \quad (2)$$

A finite right side makes the nonnegative tails converge to zero in L^1 ; conversely, L^1 convergence bounds the expectations of the partial sums, so (2) forces summability.

Finally, the characteristic function of the full centered partial sum is

$$\varphi_{S_J}(t) = \prod_{j \leq J} e^{-it\lambda_j} (1 - 2it\lambda_j)^{-1/2}, \quad |\varphi_{S_J}(t)| = \prod_{j \leq J} (1 + 4t^2\lambda_j^2)^{-1/4}. \quad (3)$$

If $m_J = |\{j \leq J : \lambda_j = \lambda_o\}|$, then $m_J \rightarrow \infty$ and, for every $t \neq 0$,

$$|\varphi_{S_J}(t)| \leq (1 + 4t^2\lambda_o^2)^{-m_J/4} \rightarrow 0. \quad (4)$$

If a subsequence converged weakly to a real variable S , its characteristic functions would converge to $E[e^{itS}]$, which by (4) would vanish for every $t \neq 0$. This contradicts continuity at zero of every characteristic function. Hence no weakly convergent subsequence exists. $\square$

Justification. The full centered partial sums are controlled by their actual characteristic-function product, not by a selected subseries. *Gap.* A selected-subseries argument does not by itself exclude convergence of the full partial sums. *Consumer.* It supplies the exact spectral compactness criterion and the full-partial-sum obstruction used by the degeneracy theorem.

Lemma 9 (lem:conditional-hessian-spectral-obstruction). *Suppose the variance-unrestricted parent class contains an equality law P° for which P_X° is nonatomic, $\pi^\circ(x) = 1/2$, and $q_0^\circ(x, y) = q_1^\circ(x, y) = q^\circ$ is a constant density lying in the interior of the frozen density bounds and Hölder ball. Then the equality Hessian of θ , viewed on the closure of regular observed-data scores in $L_0^2(P^\circ)$, is not compact. More precisely, there is an infinite-dimensional closed score subspace $\mathcal{E}$ and a constant $\lambda_o > 0$ such that the compression to $\mathcal{E}$ of any bounded self-adjoint Riesz representative H° of the second derivative equals $\lambda_o I_{\mathcal{E}}$. Consequently H° is neither compact nor Hilbert–Schmidt, and no unregularized square-summable spectral representation exists at this law.*

Proof. Choose nonconstant $w \in C_c^\infty(\text{int } \mathcal{Y})$ and put

$$g(y) = -q^\circ \Delta w(y), \quad \kappa = q^\circ \int_{\mathcal{Y}} \|\nabla w(y)\|_2^2 d\ell_{\mathcal{Y}}(y) > 0. \quad (1)$$

Then $\int g = 0$ and the zero-flux boundary condition holds. For bounded smooth e on $\mathcal{X}$, perturb only arm one by $q_{0,t} = q^\circ$ and $q_{1,t}(x, y) = q^\circ + te(x)g(y)$. The interior condition keeps this path in the model for sufficiently small $|t|$. Its score is

$$s_e(x, a, y) = 1\{a = 1\} \frac{e(x)g(y)}{q^\circ}. \quad (2)$$

Since $\pi^\circ = 1/2$,

$$\langle s_e, s_f \rangle_{L^2(P^\circ)} = c_g \langle e, f \rangle_{L^2(P_X^\circ)}, \quad c_g = \frac{1}{2} \int \frac{g(y)^2}{q^\circ} d\ell_Y(y) > 0. \quad (3)$$

The Neumann solution for direction $e(x)g(y)$ is $e(x)w(y)$. Polarizing the energy formula in `lem:equality-degeneracy-and-second-variation` yields

$$D^2\theta_{P^\circ}[s_e, s_f] = 2\kappa \langle e, f \rangle_{L^2(P_X^\circ)} = \lambda_\circ \langle s_e, s_f \rangle_{L^2(P^\circ)}, \quad \lambda_\circ = 2\kappa/c_g > 0. \quad (4)$$

The full outcome-score Hessian is bounded. Indeed, if $-q^\circ \Delta v_x = d(x, \cdot)$ with zero flux and mean zero, integration by parts and Poincaré give

$$q^\circ \|\nabla v_x\|_2^2 = \int d(x, y) v_x(y) d\ell_Y(y) \leq C_Y \|d(x, \cdot)\|_2 \|\nabla v_x\|_2. \quad (5)$$

Fixed overlap and the density bounds control the integrated energy by the squared observed-score norm. Polarization and the Riesz representation theorem therefore supply a bounded self-adjoint representative H° .

Because P_X° is nonatomic, $L^2(P_X^\circ)$ is infinite dimensional, and bounded smooth functions are dense. Hence $\mathcal{E} = \{s_e : e \text{ bounded and smooth}\}$ is infinite dimensional. Continuity extends (4) to $\mathcal{E}$, giving

$$\Pi_{\mathcal{E}} H^\circ|_{\mathcal{E}} = \lambda_\circ I_{\mathcal{E}}. \quad (6)$$

If H° were compact, the left side of (6) would be compact, but a positive multiple of the identity on an infinite-dimensional space is not compact. Thus H° is not compact. Every Hilbert–Schmidt operator is compact, so it is not Hilbert–Schmidt and cannot possess an unregularized square-summable spectral representation. $\square$

Lemma 10 (`lem:two-sample-quadratic-remainder`). *Fix x in the common full-measure set from the three uniform conditional Brenier assumptions. Let $\bar{P} = (\bar{P}_0, \bar{P}_1)$ and $P = (P_0, P_1)$ be probability-law pairs on the convex set $\mathcal{Y}$ such that every component has a density equal to the x -section of an element of $\mathfrak{F}$ in `def:coupled-fold-learners`; hence all four densities are normalized, belong to the common Hölder class, and lie between c and C . Assume that for every ordered pair generated by these four densities, the corresponding quadratic Brenier potential belongs to $C^2(\mathcal{Y})$ and has Hessian between $m_{\text{OT}} I_{d_Y}$ and $M_{\text{OT}} I_{d_Y}$. Let $(\bar{u}_0, \bar{u}_1)$ be a mutually conjugate optimal quadratic dual pair at $\bar{P}$, and define*

$$\mathcal{D}_{\text{OT}}(P, \bar{P}) = W_2^2(P_0, P_1) - W_2^2(\bar{P}_0, \bar{P}_1) - \sum_{a \in \mathcal{A}} \int \bar{u}_a d(P_a - \bar{P}_a).$$

There is a finite constant C_{OT} , depending only on $c, C, m_{\text{OT}}, M_{\text{OT}}$ and the geometry of $\mathcal{Y}$, such that

$$0 \leq \mathcal{D}_{\text{OT}}(P, \bar{P}) \leq C_{\text{OT}} \{W_2(P_0, \bar{P}_0) + W_2(P_1, \bar{P}_1)\}^2.$$

The assertion is unchanged by centering the base potentials.

Proof. Write $F(P) = W_2^2(P_0, P_1)$, $d_a = W_2(P_a, \bar{P}_a)$, and let (u_0, u_1) be an optimal dual pair at P . Feasibility of the base optimizer at P gives $\mathcal{D}_{\text{OT}}(P, \bar{P}) \geq 0$. Feasibility of the perturbed optimizer at $\bar{P}$, subtracted from optimality at P , gives

$$\mathcal{D}_{\text{OT}}(P, \bar{P}) \leq \sum_a \int (u_a - \bar{u}_a) d(P_a - \bar{P}_a). \quad (1)$$

The forward and reverse conclusions of `lem:two-sample-map-stability`, the map-potential identities, and the density lower bound imply

$$\sum_a \|\nabla(u_a - \bar{u}_a)\|_{L^2(\ell_Y)}^2 \leq C_1(d_0 + d_1)^2. \quad (2)$$

For each arm, let γ_a be an optimal coupling of $\bar{P}_a$ and P_a . Displacement interpolation remains in convex $\mathcal{Y}$, and the endpoint density upper bound implies that every interpolating density is at most C . Therefore, for every $v \in H^1(\mathcal{Y})$, the fundamental theorem of calculus along transport segments and Cauchy–Schwarz give

$$\left| \int v d(P_a - \bar{P}_a) \right| \leq C^{1/2} d_a \|\nabla v\|_{L^2(\ell_{\mathcal{Y}})}. \quad (3)$$

Apply (3) with $v = u_a - \bar{u}_a$, sum over arms, and use (1)–(2). This yields $0 \leq \mathcal{D}_{\text{OT}}(P, \bar{P}) \leq C_{\text{OT}}(d_0 + d_1)^2$. Adding constants to the base potentials integrates to zero against $P_a - \bar{P}_a$, so centering leaves the assertion unchanged. $\square$

Lemma 11 (lem:two-sample-map-stability). *Fix x in the common full-measure set from `ass : uniform – conditional – brenier – smoothness`, `ass : uniform – conditional – brenier – lower – curvature`, and `ass : uniform – conditional – brenier – upper – curvature`. Let $P, Q, \hat{P}, \hat{Q}$ be probability laws on the convex set $\mathcal{Y}$, each having a density that is the x -section of an element of the class $\mathfrak{F}$ in `def:coupled-fold-learners`; in particular, all four densities are normalized, belong to the common Hölder class, and lie between c and C . Let $b \in C^2(\mathcal{Y})$ be a quadratic Brenier potential from P to Q satisfying*

$$m_{\text{OT}} I_{d_Y} \preceq \nabla^2 b(y) \preceq M_{\text{OT}} I_{d_Y} \quad (y \in \mathcal{Y}),$$

put $T = \nabla b$, and let $\hat{\gamma} \in \Gamma(\hat{P}, \hat{Q})$ be an optimal quadratic coupling. Then

$$\frac{1}{M_{\text{OT}}} \int \|\zeta - T(\xi)\|_2^2 d\hat{\gamma}(\xi, \zeta) \leq \frac{1}{m_{\text{OT}}} W_2^2(Q, \hat{Q}) + M_{\text{OT}} W_2^2(P, \hat{P}) + 2W_2(P, \hat{P})W_2(Q, \hat{Q}).$$

The analogous assertion holds for the reverse maps after swapping the marginals, provided the corresponding reverse Brenier potential is in $C^2(\mathcal{Y})$ and satisfies the same all-feasible-pairs lower- and upper-curvature bounds.

Proof. Use the half-squared-cost semi-dual

$$S_{\hat{P}, \hat{Q}}(b) = \int b d\hat{P} + \int b^* d\hat{Q}, \quad \Delta = S_{\hat{P}, \hat{Q}}(b) - \inf_{\tilde{b}} S_{\hat{P}, \hat{Q}}(\tilde{b}).$$

Because b^* is M_{OT}^{-1} -strongly convex and $T = \nabla b$, Fenchel equality on an optimal fitted coupling gives

$$\Delta \geq \frac{1}{2M_{\text{OT}}} \int \|\zeta - T(\xi)\|_2^2 d\hat{\gamma}(\xi, \zeta). \quad (1)$$

For the upper bound, draw $X \sim P$, put $Y = T(X)$, couple X optimally to $U \sim \hat{P}$, and Y optimally to $V \sim \hat{Q}$. The resulting law of (U, V) is feasible. Taylor’s inequalities from $\nabla^2 b \preceq M_{\text{OT}} I$ and $\nabla^2 b^* \preceq m_{\text{OT}}^{-1} I$ give

$$b(U) + b^*(V) - \langle U, V \rangle \leq \frac{M_{\text{OT}}}{2} \|U - X\|_2^2 + \frac{1}{2m_{\text{OT}}} \|V - Y\|_2^2 - \langle U - X, V - Y \rangle.$$

Taking expectations and applying Cauchy–Schwarz yields

$$\Delta \leq \frac{M_{\text{OT}}}{2} W_2^2(P, \hat{P}) + \frac{1}{2m_{\text{OT}}} W_2^2(Q, \hat{Q}) + W_2(P, \hat{P})W_2(Q, \hat{Q}). \quad (2)$$

Combining (1)–(2) and multiplying by two proves the displayed inequality. Repeating the argument with the conjugate potential and swapped marginals proves the reverse-map assertion. $\square$

Lemma 12 (lem:assignment-score-rates). *Let $\mathcal{C} \subseteq \mathcal{P}^{\text{vu}}$ be predetermined, and assume that the generic propensity learner-rate condition in `def:propensity-rate-regimes` holds uniformly over $\mathcal{C}$. Under the conclusions of `lem:conditional-transport-stability-rates`, uniformly over $\mathbf{P} \in \mathcal{C}$,*

$$\max_k \left| E_{P_{X,n}} \sum_a \left(\frac{\pi_{a,n}}{\bar{\pi}_{a,-k,n}} - 1 \right) \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}) \right| = O_{P_n}(r_{\pi,n} r_{q,n}),$$

and

$$\max_k \left\| \hat{S}_{-k,n} - (\theta_n + \psi_n) \right\|_{L^2(P_n)} = O_{P_n}(r_{\pi,n} + 2r_{q,n}).$$

Proof. Fix a fold and suppress indices. Define $g_a(x) = \int \hat{\phi}_a(x, y) d(Q_a - \hat{Q}_a)(y | x)$. Exact fitted-kernel centering and true centering imply

$$g_a(x) = \int (\hat{\phi}_a - \phi_a)(x, y) Q_a(dy | x).$$

Jensen's inequality and `lem:conditional-transport-stability-rates` give $\|g_a\|_{L^2(P_X)} = O_{P_n}(r_{q,n})$. Fixed overlap and clipping give

$$\left| \frac{\pi_a}{\bar{\pi}_a} - 1 \right| \leq \varepsilon^{-1} |\pi_a - \bar{\pi}_a|.$$

Cauchy–Schwarz and the generic propensity premise therefore prove the first displayed product order.

For the score rate, let $S^\circ = \theta + \psi$. Then

$$\hat{S} - S^\circ = (\hat{V} - V) + \sum_a 1\{A = a\} \left\{ \frac{\hat{\phi}_a - \phi_a}{\bar{\pi}_a} + \phi_a \left(\frac{1}{\bar{\pi}_a} - \frac{1}{\pi_a} \right) \right\}. \quad (1)$$

The compact-cost conjugacy bound gives $|\phi_a| \leq 2 \text{diam}(\mathcal{Y})^2$. Taking $L^2(P)$ norms in (1), using overlap, the transport-value and potential rates, and the propensity premise, yields

$$\|\hat{S} - S^\circ\|_{L^2(P)} = O_{P_n}(r_{q,n} + r_{\pi,n}).$$

The harmless notation $r_{\pi,n} + 2r_{q,n}$ records the two transport arms. Taking a maximum over fixed K proves the assertion uniformly. $\square$

Justification. This new lemma places the generic propensity learner premise exactly where it is used and proves both the propensity–density product bias and the full coupled-score $L^2(P_n)$ rate. *Gap.* The transport-only stability conclusions do not logically imply assignment-score rates without an independently stated propensity rate. *Consumer.* It is the propensity-dependent bridge from transport stability to the cross-fit remainder and variance bounds.

Lemma 13 (`lem:conditional-transport-stability-rates`). *Uniformly over $\mathbf{P} \in \mathcal{P}^{\text{vu}}$, the conditional-density rate $r_{q,n}$ implies*

$$\max_{k,a} \int W_2^2(\hat{Q}_{a,-k,n}, Q_{a,P_n}) dP_{X,n} = O_{P_n}(r_{q,n}^2),$$

$$\max_{k,a} \|\hat{\phi}_{a,-k,n} - \phi_{a,n}\|_{L^2(\nu_{a,n})} = O_{P_n}(r_{q,n}), \quad \max_k \|\hat{V}_{-k,n} - V_n\|_{L^2(P_{X,n})} = O_{P_n}(r_{q,n}).$$

With the fitted pair as base and the truth as perturbation,

$$\hat{V}_{-k,n} - V_n + \sum_a \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}) = -\mathcal{D}_{\text{OT}}(Q, \hat{Q}) \leq 0,$$

and, explicitly,

$$\max_k \left\| \hat{V}_{-k,n} - V_n + \sum_a \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}) \right\|_{L^1(P_{X,n})} = \max_k E_{P_{X,n}} \mathcal{D}_{\text{OT}}(Q, \hat{Q}) = O_{P_n}(r_{q,n}^2).$$

Proof. Fix a fold and suppress indices. Put $e_a = \hat{q}_a - q_a$ and $d_a(x) = W_2(\hat{Q}_a(\cdot | x), Q_a(\cdot | x))$. By `lem:wasserstein-l2-density-control`,

$$d_a(x)^2 \leq c^{-1} C_Y^2 \int e_a(x, y)^2 d\ell_Y(y). \quad (1)$$

Integrating and invoking `lem:finite-net-conditional-density-rate` proves the first assertion.

At a fixed admissible x , write $h_a = \hat{u}_a - u_a$. Applying the forward and reverse forms of `lem:two-sample-map-stability`, using the map–potential gradient identities and the density lower bound, gives

$$\sum_a \|\nabla_y h_a(x, \cdot)\|_{L^2(\ell_Y)}^2 \leq C_1 \{d_0(x) + d_1(x)\}^2. \quad (2)$$

Exact centering gives

$$\hat{\phi}_a - \phi_a = h_a - Q_a h_a + (Q_a - \hat{Q}_a) \hat{u}_a. \quad (3)$$

Weighted Poincaré controls the first two terms in (3) by the gradient in (2). The common oscillation bound for quadratic dual potentials and Cauchy–Schwarz give

$$|(Q_a - \hat{Q}_a) \hat{u}_a| \leq C_2 \|e_a(x, \cdot)\|_{L^2(\ell_Y)}. \quad (4)$$

Squaring, integrating, and using the density rate proves the potential rate. If $w = W_2(Q_0, Q_1)$ and $\hat{w} = W_2(\hat{Q}_0, \hat{Q}_1)$, the metric triangle inequality gives $|\hat{w} - w| \leq d_0 + d_1$. Since all distances are at most $\text{diam}(\mathcal{Y})$,

$$|\hat{V} - V| \leq 2 \text{diam}(\mathcal{Y})(d_0 + d_1), \quad (5)$$

which proves the value rate.

Finally apply `lem:two-sample-quadratic-remainder`, pointwise in x , with the fitted pair as base and truth as perturbation. Centering does not alter the linear term, so

$$\hat{V} - V + \sum_a \int \hat{\phi}_a d(Q_a - \hat{Q}_a) = -\mathcal{D}_{\text{OT}}(Q, \hat{Q}) \leq 0. \quad (6)$$

Moreover $E_{P_X} \mathcal{D}_{\text{OT}} \leq C E_{P_X} (d_0 + d_1)^2 = O_{P_n}(r_{q,n}^2)$. Because the left side of (6) is nonpositive, its $L^1(P_X)$ -norm is exactly that expectation. Taking maxima over the fixed folds proves all conclusions uniformly. $\square$

Lemma 14 (`lem:regular-analytic-gradient-conditions`). *For every $P_n \in \mathcal{M}_n^{\text{reg}}$ satisfying `ass : regular – support – geometry`, the unique-normalized-potential property $\text{U}(P_n)$ in `def:unique-normalized-conditional-potentials` and the square-integrable-gradient property $\text{G}(P_n)$ in `def:gradient-square-integrability` hold.*

Proof. Fix P_n and suppress n . For almost every x , both conditional laws have densities bounded below on compact convex $\mathcal{Y}$. For the continuous quadratic cost, successive quadratic transforms of a feasible maximizing sequence, normalized by $\min u_0 = 0$, are uniformly bounded and equicontinuous. Arzelà–Ascoli gives a uniformly convergent subsequence; feasibility and the dual objective pass to the limit, proving existence of a normalized mutually conjugate optimal pair.

Each potential is Lipschitz and differentiable almost everywhere. Positivity of the source density and differentiation of the contact identity show that every optimal plan is concentrated on the uniquely determined graph associated with ∇u_0 . The average of two optimal graph plans is optimal, and hence must again lie on one graph; therefore their maps agree almost surely. Two normalized optimal first potentials consequently have equal gradients almost everywhere. Their Lipschitz difference has zero weak gradient on connected $\mathcal{Y}$, so it is constant, and normalization makes it zero. Conjugacy then uniquely determines the second potential.

For joint measurability, place the normalized equibounded, equicontinuous pairs in the compact uniform-metric space. Feasibility is testable on countable dense subsets, and the dual objective is measurable in x . The same nested-first-ball construction as in `lem:conditional-coupling-sharpness` supplies a measurable selector; uniqueness makes it the normalized pair just constructed. Hence $\text{U}(P_n)$ holds. Finally, `lem:uniform-canonical-gradient-bound` gives $E\psi_n = 0$ and $|\psi_n| \leq M_\psi$, so $\psi_n \in L_0^2(P_n)$, which is $\text{G}(P_n)$. $\square$

Lemma 15 (`lem:finite-net-propensity-rate`). *Let $\mathcal{P}_{\sigma_*, t_\pi, L_\pi}^{\text{vs}, \text{H}}$ be the structural Hölder-propensity subclass in `def:holder-propensity-triangular-law-class`. On every training complement use the clipped finite-net least-squares learner specified in `def:propensity-rate-regimes`. Then, uniformly over $\mathbf{P} \in \mathcal{P}_{\sigma_*, t_\pi, L_\pi}^{\text{vs}, \text{H}}$,*

$$\max_{1 \leq k \leq K} \|\bar{\pi}_{-k,n} - \pi_n\|_{L^2(P_{X,n})} = O_{P_n} \left(n^{-t_\pi / (2t_\pi + d_X)} \right).$$

Proof. Fix a fold with training size $m \asymp n$, and write $\ell_g(X, A) = (A - g(X))^2$. Conditional expectation gives the exact excess-risk identity

$$\mathcal{R}(g) = P(\ell_g - \ell_\pi) = \|g - \pi\|_{L^2(P_X)}^2. \quad (1)$$

Writing $d_g = g - \pi$, one has $\ell_g - \ell_\pi = d_g(X)^2 - 2\{A - \pi(X)\}d_g(X)$. Since $g, \pi \in [\varepsilon, 1 - \varepsilon]$, the loss difference is bounded and

$$E[(\ell_g - \ell_\pi)^2 | X] \leq 4d_g(X)^2. \quad (2)$$

Bernstein's inequality, Young's inequality, and a union bound over the finite net and fixed folds imply that, with probability at least $1 - 2Ke^{-u}$,

$$|(\mathbb{P}_{k,n} - P)(\ell_g - \ell_\pi)| \leq \frac{1}{4}\mathcal{R}(g) + C_1 \frac{\log(2|\mathfrak{G}_J|) + u}{m} \quad (3)$$

simultaneously for all candidates. The Holder assumption and net construction provide g° with $\|g^\circ - \pi\|_\infty \lesssim 2^{-Jt_\pi}$ and $\log |\mathfrak{G}_J| \lesssim 2^{Jd_X}$. Comparing the empirical minimizer to g° in (3) yields

$$\mathcal{R}(\tilde{\pi}_{-k,n}) \lesssim 2^{-2Jt_\pi} + \frac{2^{Jd_X} + u}{n}. \quad (4)$$

Projection onto $[\varepsilon, 1-\varepsilon]$ is one-Lipschitz and fixes π , so clipping cannot increase this error. With $2^{Jd_X} \asymp n^{d_X/(2t_\pi+d_X)}$, taking square roots in (4) gives the claimed $n^{-t_\pi/(2t_\pi+d_X)}$ order uniformly over laws and folds. $\square$

Theorem 1 (thm:abstract-observed-canonical-gradient). *Let P_n be an observed-data law satisfying the three analytic properties $\mathbf{U}(P_n)$, $\mathbf{D}(P_n)$, and $\mathbf{G}(P_n)$ in `def:unique-normalized-conditional-potentials`, `def:conditional-transport-pathwise-differentiability`, and `def:gradient-square-integrability`. Then ψ_n is the canonical gradient of θ_n in the nonparametric observed-data model, and $E_{P_n}[\psi_n(O)^2] = \sigma_n^2$ is the semiparametric efficiency bound.*

Proof. Fix n , suppress the row index, and write $P = P_n$. Property $\mathbf{U}(P)$ makes the normalized conditional dual potentials and their centered versions ϕ_a well defined. Let P_t be an arbitrary regular dominated one-dimensional submodel through P , with score $h \in L_0^2(P)$. Put $h_X(x) = E[h(O) \mid X = x]$, $m_a(x) = E[h(O) \mid A = a, X = x]$, and $h_{Y,a}(x, y) = h(x, a, y) - m_a(x)$. By $\mathbf{D}(P)$,

$$\dot{\theta}_P(h) = E[(V(X) - \theta)h_X(X)] + \sum_a E_{P_X} \left[\int \phi_a(X, y) h_{Y,a}(X, y) Q_a(dy \mid X) \right]. \quad (1)$$

Because $V(X) - \theta$ is X -measurable, the first term equals $E[(V(X) - \theta)h(O)]$. For each arm, iterated expectation and positivity give

$$E \left[\frac{1\{A = a\}}{\pi_a(X)} \phi_a(X, Y) h(O) \right] = E_{P_X} \left[\int \phi_a(X, y) h(x, a, y) Q_a(dy \mid X) \right]. \quad (2)$$

Exact centering, $\int \phi_a(x, y) Q_a(dy \mid x) = 0$, allows $h(x, a, y)$ in (2) to be replaced by $h_{Y,a}(x, y)$. Substitution into (1) yields $\dot{\theta}_P(h) = E[\psi(O)h(O)]$ for every regular score h . The same centering identities give $E[\psi] = 0$, and $\mathbf{G}(P)$ gives $\psi \in L_0^2(P)$.

It remains only to identify the tangent space. For every bounded $g \in L_0^2(P)$, the path $dP_t = (1 + tg)dP$, for $|t||g|_\infty < 1/2$, is regular with score g . Bounded mean-zero functions are dense in $L_0^2(P)$, by truncation followed by recentering. Hence the nonparametric observed-data tangent space is all of $L_0^2(P)$. Therefore ψ is the unique tangent-space representer of the derivative, namely the canonical gradient. Its squared norm is

$$\|\psi\|_{L^2(P)}^2 = E_P[\psi(O)^2] = \sigma_n^2,$$

which is the semiparametric efficiency bound. $\square$

Lemma 16 (lem:integrated-arm-path-differentiation). *Let $P_n \in \mathcal{M}_n^{\text{reg}}$ satisfy the unique-normalized-potential property $\mathbf{U}(P_n)$ in `def:unique-normalized-conditional-potentials`. Then the conditional-transport pathwise differentiability property $\mathbf{D}(P_n)$ in `def:conditional-transport-pathwise-differentiability` holds.*

Proof. Fix a row, write $P = P_n$, and let P_t be a regular dominated submodel with score h . If $L_t = dP_t/dP$, regularity gives $L_t = 1 + th + o_{L^1(P)}(t)$. Conditional expectation and quotient expansion imply

$$\frac{dP_{X,t}/dP_X - 1}{t} \rightarrow h_X \quad \text{in } L^1(P_X), \quad (1)$$

$$\int \left| \frac{dQ_{a,t}/dQ_a - 1}{t} - h_{Y,a} \right| d(P_X \otimes Q_a) \rightarrow 0. \quad (2)$$

Fixed overlap justifies the arm-conditioned change of measure in (2). In particular the conditional marginals converge in integrated total variation.

Let (u_0, u_1) be the unique normalized base pair and $(u_{0,t}, u_{1,t})$ normalized optimizers along the path. Compact quadratic conjugacy makes these families uniformly bounded and equicontinuous. Integrated total-variation convergence, compactness, and uniqueness at $t = 0$ imply

$$\sum_a \int |u_{a,t} - u_a|^2 d(P_X \otimes Q_a) \rightarrow 0. \quad (3)$$

Indeed, any violating subsequence has an almost-everywhere conditionally convergent subsubsequence; Arzelà–Ascoli supplies uniform potential limits, and passing to dual feasibility and value makes each limit a normalized base optimizer, contradicting uniqueness.

Put $V_t(x) = W_2^2(Q_{0,t}, Q_{1,t})$. Dual feasibility gives the envelope inequalities

$$\sum_a \int u_a d(Q_{a,t} - Q_a) \leq V_t - V \leq \sum_a \int u_{a,t} d(Q_{a,t} - Q_a). \quad (4)$$

Divide by t , integrate over P_X , and use (2)–(3). The two bounds have the same limit, namely

$$\sum_a E_{P_X} \left[\int \phi_a(X, y) h_{Y,a}(X, y) Q_a(dy | X) \right], \quad (5)$$

where centering is valid because $Q_a h_{Y,a} = 0$. The values are uniformly bounded, and (4) plus total-variation convergence gives $V_t - V \rightarrow 0$ in P_X -probability. Combining this with (1) shows that the product term $t^{-1} \int (V_t - V) d(P_{X,t} - P_X)$ tends to zero. Decomposing

$$\theta(P_t) - \theta(P) = \int (V_t - V) dP_X + \int V d(P_{X,t} - P_X) + \int (V_t - V) d(P_{X,t} - P_X)$$

and applying (1) and (5) yields exactly the derivative displayed in $D(P_n)$. $\square$

Theorem 2 (thm:degeneracy-frontier-and-spectral-obstruction). *Theorem (degeneracy frontier and obstruction to the unregularized full-tangent spectral route). Let P_n be a row law of the variance-unrestricted parent $\mathcal{P}^{\text{vu}}$, equivalently $P_n \in \mathcal{M}_n^{\text{reg}}$ satisfying `ass : regular – support – geometry` and the three all-feasible-pairs conditional Brenier conditions. Then $\sigma_n^2 = 0$ if and only if $Q_{0,P_n}(\cdot | x) = Q_{1,P_n}(\cdot | x)$ for $P_{X,n}$ -almost every x . At equality, $\theta_n = 0$, $\psi_n = 0$, and the oracle first-order Gaussian pivot has Kolmogorov distance $1/2$ from $N(0, 1)$. Along every smooth equality path described in `lem:equality-degeneracy-and-second-variation`, the second directional derivative is the displayed weighted Neumann energy. A centered Gaussian spectral calibration for a second-order Hessian exists in L^2 exactly when its eigenvalues are square summable, and uniform finite-rank calibration requires a uniform square-summable tail. The parent class does not imply this condition: whenever it contains the equality law in `lem:conditional-hessian-spectral-obstruction` with nonatomic P_X , the conditional Hessian has an infinite-dimensional compression equal to a positive multiple of the identity and is neither compact nor Hilbert–Schmidt. Consequently, the unregularized n -scale full-tangent spectral/Gaussian route named in `def:degeneracy-handle` cannot yield a uniformly tight equality calibration over $\mathcal{P}^{\text{vu}}$. A positive theorem using that route must impose or construct covariate-direction regularization, a uniform Hilbert–Schmidt tail, and uniform estimation of the regularized Hessian. On collections with $\sigma_n > 0$ satisfying the estimator and scale-aware variance expansion hypotheses of `lem:vanishing-variance-studentization-scale-aware`, the first-order studentized limit remains uniformly valid whenever*

$$\frac{\sqrt{n} \delta_n}{\sigma_n} \rightarrow 0, \quad \frac{r_{S,n}}{\sigma_n} \rightarrow 0, \quad \sqrt{n} \sigma_n \rightarrow \infty$$

uniformly. This is a route-specific negative result and does not rule out slower regularized or other derivative-aware confidence procedures; honest degeneracy-adaptive inference over the entire parent remains open.

Proof. Fix a row and suppress n . By `lem:equality-degeneracy-and-second-variation`,

$$\sigma^2 = 0 \iff Q_{0,P}(\cdot | x) = Q_{1,P}(\cdot | x) \text{ for } P_X\text{-almost every } x. \quad (1)$$

At equality that lemma also gives $\theta = 0$, $\psi = 0$ almost surely, and the stipulated zero-denominator first-order pivot is identically zero. Its distribution function is $1\{t \geq 0\}$, so

$$\sup_t |1\{t \geq 0\} - \Phi(t)| = \frac{1}{2}. \quad (2)$$

For every smooth equality path in that lemma,

$$\theta(P_t) = t^2 \mathcal{E} + o(t^2), \quad \mathcal{E} = \int q(x, y) \|\nabla_y w_x(y)\|_2^2 d(P_X \otimes \ell_Y)(x, y), \quad (3)$$

and hence the second directional derivative is $2\mathcal{E}$.

For a centered Gaussian quadratic calibration, `lem:gaussian-quadratic-spectral-criterion` gives L^2 existence exactly when $\sum_j \lambda_j^2 < \infty$, and for a family Λ it gives the exact truncation identity

$$\sup_{\lambda \in \Lambda} E \left[\left\{ \sum_{j>J} \lambda_j (Z_j^2 - 1) \right\}^2 \right] = 2 \sup_{\lambda \in \Lambda} \sum_{j>J} \lambda_j^2. \quad (4)$$

Thus uniform finite-rank approximation is equivalent to a uniformly vanishing square-summable tail. If the parent contains the equality witness in `lem:conditional-hessian-spectral-obstruction`, that lemma supplies an infinite-dimensional closed score subspace $\mathcal{E}$, $\lambda_o > 0$, and a bounded self-adjoint Hessian representative H° satisfying

$$\Pi_{\mathcal{E}} H^\circ|_{\mathcal{E}} = \lambda_o I_{\mathcal{E}}. \quad (5)$$

For an orthonormal sequence (e_j) in $\mathcal{E}$, the rank- J compressed centered Gaussian form is

$$\mathcal{Q}_J^\circ = \lambda_o \sum_{j=1}^J (Z_j^2 - 1). \quad (6)$$

The spectral lemma shows that (6) has no weakly convergent subsequence. Hence the finite-dimensional compressions are not tight. In particular, the unregularized full-tangent spectral/Gaussian construction in `def:degeneracy-handle` fails a necessary tightness condition at an admissible parent law and therefore cannot be uniformly tight over $\mathcal{P}^{\text{vu}}$. Equation (4) also proves that a positive theorem along that route must remove the identity compression by covariate-direction regularization, obtain uniformly vanishing Hilbert–Schmidt tails, and estimate the regularized Hessian sufficiently well to reproduce the associated quadratic law. This obstruction is confined to that route.

Finally, on any positive-variance collection satisfying `lem:vanishing-variance-studentization-scale-aware` and the three displayed relative-rate conditions, put

$$a_n = \sigma_n r_{S,n} + r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}}.$$

Then

$$\frac{a_n}{\sigma_n^2} = \frac{r_{S,n}}{\sigma_n} + \left(\frac{r_{S,n}}{\sigma_n} \right)^2 + \frac{1}{\sqrt{n}\sigma_n} + \frac{r_{S,n}/\sigma_n}{\sqrt{n}\sigma_n} \rightarrow 0. \quad (7)$$

The cited lemma therefore yields uniform first-order studentized normality, ratio-consistent studentization, positivity with probability tending to one, and uniform Wald coverage. Combining (1)–(7) proves every asserted positive and negative clause and leaves only the stated full-parent adaptive problem unresolved. $\square$

Justification. The theorem combines exact first-order degeneracy, the weighted-Neumann second variation, the Gaussian-series criterion, and an admissible non-Hilbert–Schmidt Hessian witness to rule out the unregularized full-tangent spectral route. *Gap.* The result is route-specific and does not establish impossibility of slower regularized or nonspectral confidence procedures. *Consumer.* It explains why the positive-variance Wald result cannot be extended to equality by an unregularized spectral plug-in.

Lemma 17 (`lem:cross-fit-remainder-variance-scale-aware`). *Let $\mathcal{C} \subseteq \mathcal{P}^{\text{vu}}$ be predetermined and suppose that the density and propensity learner-rate premises used by `lem:conditional-transport-stability-rates` and `lem:assignment-score-rates` hold uniformly over $\mathcal{C}$. The exact conditional fold bias is*

$$\mathcal{B}_{k,n} = -E_{P_{X,n}} \mathcal{D}_{\text{OT}}(Q, \hat{Q}) + E_{P_{X,n}} \sum_a \left(\frac{\pi_{a,n}}{\bar{\pi}_{a,-k,n}} - 1 \right) \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}).$$

The first term is nonpositive and $O_{P_n}(r_{q,n}^2)$, and the second is $O_{P_n}(r_{\pi,n}r_{q,n})$. Uniformly over $\mathbf{P} \in \mathcal{C}$,

$$\hat{\theta}_n - \theta_n = \frac{1}{n} \sum_{i=1}^n \psi_n(O_{i,n}) + O_{P_n} \left(r_{q,n}^2 + r_{\pi,n}r_{q,n} + n^{-1/2}r_{S,n} \right),$$

and the scale-aware empirical-variance bound is

$$\hat{\sigma}_n^2 - \sigma_n^2 = O_{P_n} \left(\sigma_n r_{S,n} + r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}} \right).$$

Proof. Put $m_{k,n} = |I_{k,n}|$, $w_{k,n} = m_{k,n}/n$, let $\mathbb{P}_{k,n}$ be the fold empirical measure, and define

$$\Delta_{k,n}(o) = \hat{S}_{-k,n}(o) - \{\theta_n + \psi_n(o)\}.$$

Conditional on the training complement, iterated expectation gives

$$P_n \hat{S}_{-k,n} = E_{P_{X,n}} \left[\hat{V}_{-k,n} + \sum_a \frac{\pi_{a,n}}{\bar{\pi}_{a,-k,n}} \int \hat{\phi}_{a,-k,n} dQ_{a,P_n} \right]. \quad (1)$$

Use $\int \hat{\phi}_a d\hat{Q}_a = 0$, add and subtract $\hat{Q}_a$, and write $\pi_a/\bar{\pi}_a = 1 + (\pi_a/\bar{\pi}_a - 1)$. The signed remainder identity in `lem:conditional-transport-stability-rates` and the product bound in `lem:assignment-score-rates` then yield exactly the displayed fold-bias identity, with orders $O_{P_n}(r_{q,n}^2)$ and $O_{P_n}(r_{\pi,n}r_{q,n})$.

The estimator admits the exact decomposition

$$\hat{\theta}_n - \theta_n = \frac{1}{n} \sum_i \psi_n(O_{i,n}) + \sum_k w_{k,n} P_n \Delta_{k,n} + \sum_k w_{k,n} (\mathbb{P}_{k,n} - P_n) \Delta_{k,n}. \quad (2)$$

Conditionally on training, the last term has foldwise second moment at most $P_n \Delta_{k,n}^2 / m_{k,n}$. The score-rate conclusion $\max_k \|\Delta_{k,n}\|_2 = O_{P_n}(r_{S,n})$, balanced folds, fixed K , and conditional Chebyshev therefore make that term $O_{P_n}(n^{-1/2}r_{S,n})$. Combining with the bias orders proves the estimator expansion.

Write $e_n = \hat{\theta}_n - \theta_n$. Empirical centering gives

$$\hat{\sigma}_n^2 = \sum_k w_{k,n} \mathbb{P}_{k,n} (\psi_n + \Delta_{k,n})^2 - e_n^2. \quad (3)$$

The uniform envelope $|\psi_n| \leq M_\psi$ implies $\text{Var}(\psi_n^2) \leq M_\psi^2 \sigma_n^2$, whence

$$\sum_k w_{k,n} \mathbb{P}_{k,n} \psi_n^2 - \sigma_n^2 = O_{P_n}(\sigma_n / \sqrt{n}). \quad (4)$$

Cauchy–Schwarz gives $|P_n(\psi_n \Delta_{k,n})| = O_{P_n}(\sigma_n r_{S,n})$, while conditional Chebyshev gives an empirical fluctuation $O_{P_n}(r_{S,n}/\sqrt{n})$. Conditional Markov gives $\sum_k w_{k,n} \mathbb{P}_{k,n} \Delta_{k,n}^2 = O_{P_n}(r_{S,n}^2)$. Finally, (2) implies

$$e_n = O_{P_n} \left(\frac{\sigma_n}{\sqrt{n}} + r_{S,n}^2 + \frac{r_{S,n}}{\sqrt{n}} \right), \quad e_n^2 = O_{P_n} \left(r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}} \right). \quad (5)$$

Expanding (3) and applying (4)–(5) proves the stated scale-aware variance bound uniformly. $\square$

Lemma 18 (`lem:vanishing-variance-studentization-scale-aware`). *Consider any collection of i.i.d. triangular sequences in the variance-unrestricted parent class with $\sigma_n > 0$, $E[\psi_n] = 0$, and $|\psi_n| \leq M_\psi$. Suppose, uniformly over that collection,*

$$\hat{\theta}_n - \theta_n = \frac{1}{n} \sum_{i=1}^n \psi_n(O_{i,n}) + O_{P_n}(\delta_n),$$

$$\hat{\sigma}_n^2 - \sigma_n^2 = O_{P_n} \left(\sigma_n r_{S,n} + r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}} \right),$$

where

$$\frac{\sqrt{n} \delta_n}{\sigma_n} \rightarrow 0, \quad \frac{r_{S,n}}{\sigma_n} \rightarrow 0, \quad \sqrt{n} \sigma_n \rightarrow \infty$$

uniformly. Then the studentized statistic converges uniformly to $N(0, 1)$ in Kolmogorov distance, $\hat{\sigma}_n/\sigma_n \rightarrow 1$ uniformly in probability, $P_n(\hat{\sigma}_n > 0) \rightarrow 1$ uniformly, and the associated Wald intervals have uniform asymptotic coverage $1 - \alpha$. For the frozen one-step decomposition one may take $\delta_n = r_{q,n}^2 + r_{\pi,n} r_{q,n} + n^{-1/2}(r_{\pi,n} + 2r_{q,n})$ and $r_{S,n} = r_{\pi,n} + 2r_{q,n}$.

Proof. Define $Z_n = (\sigma_n \sqrt{n})^{-1} \sum_{i=1}^n \psi_n(O_{i,n})$. For fixed t , Taylor expansion of the complex exponential, together with $E\psi_n = 0$, $E\psi_n^2 = \sigma_n^2$, and $|\psi_n|^3 \leq M_\psi \psi_n^2$, gives

$$E \exp \left(\frac{it\psi_n}{\sigma_n \sqrt{n}} \right) = 1 - \frac{t^2}{2n} + \rho_n(t), \quad |\rho_n(t)| \leq \frac{|t|^3 M_\psi}{6\sigma_n n^{3/2}}. \quad (1)$$

Since $\sqrt{n} \sigma_n \rightarrow \infty$ uniformly, rowwise independence implies $E[e^{itZ_n}] \rightarrow e^{-t^2/2}$ uniformly. A subsequence argument and continuity of Φ upgrade this to uniform Kolmogorov convergence.

Put

$$a_n = \sigma_n r_{S,n} + r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}}.$$

The scale assumptions imply

$$\frac{a_n}{\sigma_n^2} = \frac{r_{S,n}}{\sigma_n} + \left(\frac{r_{S,n}}{\sigma_n} \right)^2 + \frac{1}{\sqrt{n} \sigma_n} + \frac{r_{S,n}/\sigma_n}{\sqrt{n} \sigma_n} \rightarrow 0. \quad (2)$$

Hence the variance expansion gives $\hat{\sigma}_n^2/\sigma_n^2 \rightarrow 1$ uniformly in probability. Nonnegativity and $\sigma_n > 0$ then give $\hat{\sigma}_n/\sigma_n \rightarrow 1$ and $P_n(\hat{\sigma}_n > 0) \rightarrow 1$ uniformly. The estimator expansion and $\sqrt{n} \delta_n/\sigma_n \rightarrow 0$ give

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta_n)}{\hat{\sigma}_n} = \frac{Z_n + o_{P_n}(1)}{\hat{\sigma}_n/\sigma_n}. \quad (3)$$

Uniform Slutsky, again justified by subsequences, proves uniform standard-normal convergence. The zero-denominator convention affects only an event of uniformly vanishing probability. Evaluating (3) at $\pm z_{1-\alpha/2}$ proves uniform Wald coverage. The last displayed choices of δ_n and $r_{S,n}$ are exactly those supplied by `lem:cross-fit-remainder-variance-scale-aware`. $\square$

Theorem 3 (thm:observed-gradient-minimal). *Fix a row law P_n . If a causal extension satisfies consistency, conditional ignorability, and arm positivity $P_n\{\pi_{a,n}(X_n) > 0\} = 1$ for $a = 0, 1$, then $Q_{a,P_n}(\cdot | x) = \mu_{a,P_n}(\cdot | x)$ for $P_{X,n}$ -almost every x and every a . Consequently, θ_n is the sharp conditional-coupling lower endpoint on $E[\|Y_n(1) - Y_n(0)\|_2^2]$. Separately, at every observed law satisfying the analytic properties $\mathbf{U}(P_n)$, $\mathbf{D}(P_n)$, and $\mathbf{G}(P_n)$, ψ_n is the canonical gradient of θ_n in the nonparametric observed-data model and σ_n^2 is its semiparametric efficiency bound. Membership in $\mathcal{M}_n^{\text{reg}}$ together with `ass:regular-support-geometry` is sufficient for these three analytic properties.*

Proof. Fix a row and suppress n . On $\{A = a\}$, consistency gives $1\{Y \in B\} = 1\{Y(a) \in B\}$. Hence

$$Q_{a,P}(B | x) = P\{Y(a) \in B | A = a, X = x\}. \quad (1)$$

Ignorability changes the right side to $P\{Y(a) \in B | X = x\} = \mu_{a,P}(B | x)$, and positivity makes (1) valid for P_X -almost every x . Applying this equality to a countable determining class and then a monotone-class argument yields equality of the kernels.

Every compatible causal extension therefore has conditional potential-outcome marginals Q_0, Q_1 . By `lem:conditional-coupling-sharpness`,

$$\inf E\|Y(1) - Y(0)\|_2^2 = \int W_2^2(Q_0(\cdot | x), Q_1(\cdot | x)) dP_X(x) = \theta_n, \quad (2)$$

and the measurable optimal-coupling construction in that lemma attains (2) in a compatible causal extension. Thus the endpoint is sharp.

If $U(P_n)$, $D(P_n)$, and $G(P_n)$ hold, **thm:abstract-observed-canonical-gradient** directly identifies ψ_n as the canonical gradient and σ_n^2 as its squared norm. Finally, if $P_n \in \mathcal{M}_n^{\text{reg}}$ and the regular-support geometry holds, **lem:regular-analytic-gradient-conditions** gives $U(P_n)$ and $G(P_n)$, and **lem:integrated-arm-path-differentiation** then gives $D(P_n)$. This proves the advertised primitive sufficiency. $\square$

Justification. Consistency, conditional ignorability, and positivity identify the conditional marginal laws, measurable optimal coupling makes the lower endpoint sharp, and the analytic derivative properties yield the observed-data canonical gradient. *Gap.* This theorem identifies and differentiates the lower endpoint; it does not identify the full joint law of the two potential outcomes. *Consumer.* It separates the causal endpoint from the observed-law influence function used by every subsequent inference result.

Theorem 4 (**thm:regular-efficient-inference-scale-aware**). *Theorem (conditional observational DML for the sharp conditional-optimal-transport endpoint with positive vanishing variance). Let $\mathcal{C} \subseteq \mathcal{P}^{\text{vu}}$ be any predetermined collection such that $\sigma_n > 0$ for every row law under consideration. Use the predetermined normalized coupled learners in **def:coupled-fold-learners** and suppose their uniform conditional-density rate premise holds over $\mathcal{C}$. Use a predetermined propensity learner satisfying the generic uniform $L^2(P_{X,n})$ rate premise in **def:propensity-rate-regimes**, for deterministic $r_{\pi,n} \rightarrow 0$. Put*

$$\delta_n = r_{q,n}^2 + r_{\pi,n}r_{q,n} + n^{-1/2}(r_{\pi,n} + 2r_{q,n}), \quad r_{S,n} = r_{\pi,n} + 2r_{q,n}.$$

Suppose, uniformly over $\mathcal{C}$,

$$\frac{\sqrt{n}\delta_n}{\sigma_n} \rightarrow 0, \quad \frac{r_{S,n}}{\sigma_n} \rightarrow 0, \quad \sqrt{n}\sigma_n \rightarrow \infty.$$

Then the exact fold bias is

$$\mathcal{B}_{k,n} = -E_{P_{X,n}} \mathcal{D}_{\text{OT}}(Q, \hat{Q}) + E_{P_{X,n}} \sum_a \left(\frac{\pi_{a,n}}{\bar{\pi}_{a,-k,n}} - 1 \right) \int \hat{\phi}_{a,-k,n} d(Q_{a,P_n} - \hat{Q}_{a,-k,n}),$$

where $\mathcal{D}_{\text{OT}} \geq 0$, and its two terms are respectively $O_{P_n}(r_{q,n}^2)$ and $O_{P_n}(r_{\pi,n}r_{q,n})$. Moreover,

$$\hat{\theta}_n - \theta_n = \frac{1}{n} \sum_{i=1}^n \psi_n(O_{i,n}) + R_n, \quad R_n = O_{P_n}(\delta_n) = o_{P_n}(\sigma_n/\sqrt{n}),$$

$$\hat{\sigma}_n^2 - \sigma_n^2 = O_{P_n} \left(\sigma_n r_{S,n} + r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}} \right) = o_{P_n}(\sigma_n^2).$$

The studentized statistic converges uniformly to $N(0,1)$ in Kolmogorov distance, $\hat{\sigma}_n/\sigma_n \rightarrow 1$ uniformly, $P_n(\hat{\sigma}_n > 0) \rightarrow 1$ uniformly, and the two-sided Wald interval has uniform asymptotic coverage $1 - \alpha$. The influence function is the observed-data canonical gradient ψ_n , so the leading variance attains the semi-parametric efficiency bound. Unlike the unconditional one- or two-marginal-sample result summarized in **lem:manole-unconditional-smooth-wasserstein-efficiency**, this observational-sample theorem includes the covariate-composition and inverse-propensity terms in ψ_n .

Proof. Fix $\mathcal{C}$. The density and propensity premises permit application of **lem:conditional-transport-stability-rates**, **lem:assignment-score-rates**, and then **lem:cross-fit-remainder-variance-scale-aware**. These lemmas give the exact displayed fold-bias identity; its transport remainder is nonpositive and $O_{P_n}(r_{q,n}^2)$, while its assignment term is $O_{P_n}(r_{\pi,n}r_{q,n})$. They also give

$$\hat{\theta}_n - \theta_n = \frac{1}{n} \sum_i \psi_n(O_{i,n}) + R_n, \quad R_n = O_{P_n}(\delta_n), \quad (1)$$

$$\hat{\sigma}_n^2 - \sigma_n^2 = O_{P_n} \left(\sigma_n r_{S,n} + r_{S,n}^2 + \frac{\sigma_n + r_{S,n}}{\sqrt{n}} \right). \quad (2)$$

The first relative-rate assumption turns (1) into $R_n = o_{P_n}(\sigma_n/\sqrt{n})$. Dividing the deterministic envelope in (2) by σ_n^2 gives

$$\frac{r_{S,n}}{\sigma_n} + \left(\frac{r_{S,n}}{\sigma_n} \right)^2 + \frac{1}{\sqrt{n}\sigma_n} + \frac{r_{S,n}/\sigma_n}{\sqrt{n}\sigma_n} \rightarrow 0, \quad (3)$$

by the other two relative-rate assumptions. Hence (2) is $o_{P_n}(\sigma_n^2)$ uniformly.

The canonical gradients are centered and uniformly bounded by **lem:uniform-canonical-gradient-bound**. All hypotheses of **lem:vanishing-variance-studentization-scale-aware** therefore hold. It yields uniform Kolmogorov convergence of the studentized statistic to $N(0, 1)$, uniform ratio consistency of $\hat{\sigma}_n$, positivity with probability tending uniformly to one, and Wald coverage $1 - \alpha$. Finally, **thm:observed-gradient-minimal** identifies ψ_n as the canonical gradient and σ_n^2 as the efficiency bound, so (1) attains that bound. The comparison with unconditional marginal sampling follows from the already-stated **lem:manole-unconditional-smooth-wasserstein-efficiency**; the additional composition and inverse-propensity terms are explicit in ψ_n . $\square$

Justification. Lin–Gao–Blanchet–Glynn already establish smooth-density wavelet conditional-optimal-transport stability and randomized marginal-sample asymptotic normality. The present theorem adds three distinct observational conclusions: the covariate-composition plus inverse-propensity canonical gradient under ignorability and overlap; the semiparametric information lower bound together with an attaining cross-fitted one-step estimator; and uniform studentized normality and Wald coverage on admissible positive collections with $\sigma_n \rightarrow 0$ under the displayed scale-aware conditions. *Gap.* The theorem does not cover exact conditional equality, where the first-order influence function vanishes, and it does not claim that its sufficient nuisance-rate or vanishing-variance boundaries are minimax necessary. *Consumer.* It supports efficiency-attaining, propensity-corrected Wald inference for the sharp 401(k) SIPP endpoint when ignorability, overlap, transport regularity, learner rates, and the positive-variance scale conditions are credible; Ji–Lei–Spector’s dual-bound AIPW procedure remains the distinct observational comparator.

Proposition 1 (prop:holder-propensity-efficient-inference-phase-boundary). *Put $a_q = s/(2s + D)$ and $a_\pi = t_\pi/(2t_\pi + d_X)$. Over $\mathcal{P}_{\sigma_*, t_\pi, L_\pi}^{\text{vs}, \text{H}}$, the finite-net learners satisfy the hypotheses of **thm:regular-efficient-inference-scale-aware** whenever $s > D/2$ and $a_\pi + a_q > 1/2$, equivalently $t_\pi > Dd_X/(4s)$. More generally, on any predetermined positive-variance collection with these learner rates and constants $0 < c_\sigma \leq C_\sigma < \infty$ such that $c_\sigma n^{-\kappa} \leq \sigma_n \leq C_\sigma n^{-\kappa}$ uniformly for some $\kappa \geq 0$, the scale-aware theorem applies whenever*

$$\kappa < \min \left\{ 2a_q - \frac{1}{2}, a_\pi + a_q - \frac{1}{2}, a_q, a_\pi, \frac{1}{2} \right\}.$$

If the propensity is known, the a_π -dependent terms are absent, so the boundary is $\kappa < \min\{2a_q - 1/2, a_q, 1/2\}$. If a correctly specified fixed-dimensional propensity learner has uniform rate $n^{-1/2}$, the same boundary results after setting $a_\pi = 1/2$. These sufficient thresholds are specific to the present L^2 -conditional-density and first-order remainder route; no minimax necessity is claimed.

Proof. By **lem:finite-net-propensity-rate** and **def:foldwise-rate-route**, $r_{\pi,n} = n^{-a_\pi}$, $r_{q,n} = n^{-a_q}$, and $r_{S,n} = n^{-a_\pi} + 2n^{-a_q}$. If $\sigma_n \geq \sigma_*$, the nonautomatic conditions in the scale-aware theorem are $\sqrt{nr_{q,n}^2} \rightarrow 0$ and $\sqrt{nr_{\pi,n}r_{q,n}} \rightarrow 0$. These are $2a_q > 1/2$ and $a_\pi + a_q > 1/2$. The first is equivalent to $s > D/2$, while direct cross multiplication gives

$$\frac{t_\pi}{2t_\pi + d_X} + \frac{s}{2s + D} > \frac{1}{2} \iff 4st_\pi > Dd_X \iff t_\pi > \frac{Dd_X}{4s}. \quad (1)$$

Now suppose $\sigma_n \asymp n^{-\kappa}$. The four components of $\sqrt{n}\delta_n/\sigma_n$ have orders

$$n^{\kappa+1/2-2a_q}, \quad n^{\kappa+1/2-a_\pi-a_q}, \quad n^{\kappa-a_\pi}, \quad n^{\kappa-a_q}. \quad (2)$$

Thus they vanish under $\kappa < 2a_q - 1/2$, $\kappa < a_\pi + a_q - 1/2$, $\kappa < a_\pi$, and $\kappa < a_q$. The condition $r_{S,n}/\sigma_n \rightarrow 0$ repeats the last two, and $\sqrt{n}\sigma_n \rightarrow \infty$ adds $\kappa < 1/2$. Their conjunction is the displayed minimum. If propensity is known, delete the a_π -dependent terms. If its uniform rate is $n^{-1/2}$, set $a_\pi = 1/2$. These substitutions prove every stated boundary and make no necessity claim. $\square$

Justification. The proposition substitutes the proved finite-net density and propensity rates into the scale-aware theorem and solves the resulting exponent inequalities. *Gap.* The boundary is sufficient for this learner and remainder route and is not claimed minimax necessary. *Consumer.* It gives concrete nuisance-smoothness and near-degeneracy planning benchmarks.

Theorem 5 (thm:observational-efficiency-lower-bound). *At every fixed positive-variance observed law satisfying $\mathbf{U}(P_n)$, $\mathbf{D}(P_n)$, and $\mathbf{G}(P_n)$, every regular asymptotically linear estimator of θ_n has asymptotic variance at least σ_n^2 . Along every triangular sequence satisfying the hypotheses of **thm:regular-efficient-inference-scale-aware**, the proposed cross-fitted one-step estimator has influence function ψ_n and attains this bound. Hence the efficiency statement is two-sided: σ_n^2 is both an information lower bound and the achieved leading variance.*

Proof. At a fixed law satisfying $\mathbf{U}(P_n)$, $\mathbf{D}(P_n)$, and $\mathbf{G}(P_n)$, **thm:observed-gradient-minimal** identifies ψ_n as the canonical gradient and gives $\|\psi_n\|_{L^2(P_n)}^2 = \sigma_n^2$. By **lem:canonical-gradient-information-inequality**, every regular asymptotically linear estimator therefore has asymptotic variance at least σ_n^2 .

Along a triangular sequence satisfying **thm:regular-efficient-inference-scale-aware**, that theorem gives

$$\hat{\theta}_n - \theta_n = \frac{1}{n} \sum_{i=1}^n \psi_n(O_{i,n}) + o_{P_n}(\sigma_n/\sqrt{n}).$$

Thus the proposed estimator has influence function ψ_n and leading variance σ_n^2/n , or normalized asymptotic variance σ_n^2 . It attains the information lower bound, proving both directions of the efficiency statement. $\square$

Justification. The canonical-gradient information inequality matches the leading variance achieved by the cross-fitted estimator, making the efficiency statement two-sided. *Gap.* The information bound is first-order and does not calibrate inference at equality. *Consumer.* It certifies that no regular asymptotically linear estimator improves the first-order variance at a fixed positive-variance regular law.

7 Interpretation

The causal target is the sharp lower endpoint θ_n , not the generally unidentified full moment $E[\|Y_n(1) - Y_n(0)\|_2^2]$. Its observed-data canonical gradient has a covariate-composition term and one inverse-propensity centered transport derivative for the realized arm. This is the observational ingredient beyond randomized marginal-sample smooth COT inference. On a predetermined positive-variance collection satisfying the learner premises and the three relative-rate conditions, $\hat{\theta}_n$ is asymptotically linear with this influence function, attains the information bound, $\hat{\sigma}_n/\sigma_n \rightarrow 1$, and $[\hat{\theta}_n \pm z_{1-\alpha/2}\hat{\sigma}_n/\sqrt{n}]$ has uniform limiting coverage $1 - \alpha$, including admissible sequences with $\sigma_n \rightarrow 0$. In a 401(k) SIPP analysis, these conclusions require credible ignorability and overlap, transport regularity and nuisance learning sufficient for the displayed rates, and positive row variances; exact conditional equality requires a different calibration.

Technical internal limitation (diagnostic only)

The Gaussian theorem covers positive variance only. Its vanishing-variance extension requires $\sqrt{n}\delta_n/\sigma_n \rightarrow 0$, $r_{S,n}/\sigma_n \rightarrow 0$, and $\sqrt{n}\sigma_n \rightarrow \infty$; it does not include exact equality. At equality, $\theta_n = 0$, $\psi_n = 0$, and the first-order pivot is nonnormal. The second variation is a weighted Neumann energy, and at the stated nonatomic equality laws the full-tangent Hessian has an infinite-dimensional positive identity compression. This rules out uniformly tight unregularized Gaussian spectral calibration over the full parent, but it does not rule out slower covariate-regularized, derivative-aware, subsampling, or nonspectral procedures. Constructing and validating one procedure that adapts honestly through equality remains open.

8 Honest scope

Sharp causal identification uses consistency, conditional ignorability, arm positivity, and measurable conditional-coupling attainment. The canonical-gradient claim instead uses a unique measurable normalized dual derivative, conditional-transport pathwise differentiability, and square integrability; Hölder density and all-feasible-pairs curvature conditions enter only the explicit nuisance-rate route. Uniform Gaussian inference holds on predetermined

positive-variance collections satisfying the density and propensity learner premises and $\sqrt{n} \delta_n / \sigma_n \rightarrow 0$, $r_{S,n} / \sigma_n \rightarrow 0$, and $\sqrt{n} \sigma_n \rightarrow \infty$. The sufficient thresholds $s > D/2$ and $t_\pi > Dd_X/(4s)$ are not claimed minimax necessary. No claim is made under hidden confounding, overlap failure, unsupported transport irregularity, or exact equality, and the degeneracy result remains a route-specific obstruction rather than a general impossibility theorem.

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